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SURVEY

A Comprehensive Survey of Explainable Artificial Intelligence Techniques for Malicious Insider Threat Detection

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ABSTRACT Malicious insider threats remain a persistent and formidable challenge for organizations, primarily due to their covert nature and the severe impact they can have on critical systems and sensitive data. Traditional detection mechanisms often struggle to uncover such threats, underscoring the need for more intelligent, interpretable, and trustworthy solutions. Although the research community has shown increasing interest in Insider Threat Detection (ITD), existing surveys rarely emphasize the integration of Explainable Artificial Intelligence (XAI) with machine learning (ML) and deep learning (DL) techniques. This oversight has limited progress in understanding how interpretability can improve both detection effectiveness and the trust of security analysts. To address this gap, this survey presents a comprehensive review of the application of XAI in ITD. It explores how ML and DL models, when combined with XAI techniques, can uncover anomalous behaviors and insider actions while enhancing the transparency of model decisions. Tools such as SHAP and LIME are examined for their role in revealing feature contributions and improving analyst insight. The paper also highlights critical data sources–ranging from behavioral logs and network activity to psychometric indicators–that support the development of interpretable detection models. We categorize existing literature based on XAI techniques, data modalities, and threat models, and propose a conceptual framework for aligning XAI methods with specific ITD challenges. Our findings reveal that while XAI enhances interpretability and trust in AI-driven threat detection, several challenges persist. These include class imbalance in datasets, integration of heterogeneous data streams, and the absence of standardized metrics for evaluating explainability in cybersecurity contexts. Finally, the survey identifies key directions for future research, including privacy-preserving AI, human-in-the-loop explainability, and the development of benchmarking frameworks tailored to ITD applications. By offering a structured and up-to-date overview of XAI-enhanced ITD approaches, this work supports the advancement of more transparent, accountable, and operationally effective insider threat detection systems.

INDEX TERMS Explainable AI, insider threat detection, explainable cyber security, decision trees, artificial intelligence, deep learning, neural networks.

I. INTRODUCTION

In today's digitally connected enterprises, insider threats have emerged as one of the most formidable challenges in cybersecurity. Unlike external attacks that exploit vulnerabilities from outside the organizational perimeter, insider threats stem from individuals within the organization such as employees, contractors, or partners–who possess legitimate access to sensitive systems and data. These insiders, whether acting out of malice, negligence, or coercion, can cause significant harm through unauthorized data access, intellectual property theft, or disruption of critical services. According to the 2022 Verizon Data Breach Investigations Report, insider threats account for a considerable portion

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of security incidents, highlighting the pressing need for effective detection and mitigation strategies [1], [2]. Further emphasizing the gravity of the issue, IBM's 2022 report noted that the global average cost of a data breach rose to $4.45 million in 2023, a 2.3% increase over the previous year [3].

Detecting insider threats is inherently complex. These threats often involve the misuse of legitimate privileges, making them difficult to distinguish from routine user behavior. Insider threats typically involve the misuse of legitimate privileges. Nwakanma et al. [4] categorize these threats into behaviors such as unauthorized data access, privilege escalation, and sabotage. For instance, data exfiltration can occur outside regular business hours or via disposable media devices [5]. Privilege misuse may involve an authorized user abusing access rights, while privilege escalation refers to attempts by unauthorized users to obtain elevated access [6]. These behaviors are often difficult to detect, as they closely resemble legitimate user activity [7]. Figure 1 illustrates the major categories of insider threats.

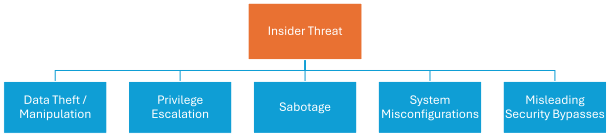


FIGURE 1. Critical insider threat categories.

To safeguard digital infrastructure against such sophisticated threats, cybersecurity systems must be designed to ensure the confidentiality, integrity, and availability of information across interconnected platforms. However, traditional signature-based and rule-based defenses are increasingly strained under the volume and complexity of modern data flows [7]. Meanwhile, threat actors–particularly malicious insiders–are adopting advanced technologies, including Artificial Intelligence (AI), to develop stealthier and more adaptive attack strategies [8]. This evolving threat landscape has driven researchers and practitioners to explore AI-based defenses, especially leveraging Machine Learning (ML) and Deep Learning (DL) models, which can analyze vast datasets and uncover patterns that would otherwise remain hidden. Yet, the increasing reliance on complex AI systems introduces new challenges, particularly in terms of transparency and interpretability [2].

Although ML and DL models perform well in cybersecurity tasks like intrusion detection and anomaly classification, they often lack transparency due to their closed-box nature. This is a major issue in high-risk areas such as insider threat detection (ITD), where explainability is critical–especially when false alarms or undetected threats can have serious consequences [2], [9]. Many models focus more on accuracy than interpretability, making them hard to audit or explain. Regulatory standards like the GDPR also highlight the need for transparent AI decisions that affect individuals [10]. To overcome these challenges, Explainable Artificial Intelligence (XAI) has become an important approach to improve trust, accountability, and transparency in cybersecurity. In ITD, XAI techniques help analysts understand why a model flagged a certain behavior as suspicious. Tools like SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) provide both global and local insights into feature importance, making it easier for security teams to verify and improve AI decisions [4], [9]. This increases the reliability and acceptance of AI-based ITD systems.

A. RESEARCH MOTIVATION

The implementation of AI in cybersecurity applications has attracted significant research interest in recent years, with numerous surveys summarizing existing work in this field. Simultaneously, the adoption of XAI has gained momentum across domains such as healthcare, finance, and image analysis–driven by the need for interpretability and transparency in AI decision-making. However, despite the availability of notable survey papers on XAI and ITD as individual topics, to the best of our knowledge, there is a noticeable gap in the literature: a comprehensive survey that specifically focuses on XAI-driven solutions across a broad spectrum of cybersecurity applications is lacking. This survey is motivated by the need to consolidate the growing body of research at the intersection of XAI and ITD. The key motivations driving this work are:

- To review and categorize prominent XAI techniques applicable to ITD.
- To identify existing XAI-based frameworks and evaluate the datasets commonly used in ITD research.
- To review recent advancements and successful implementations of XAI-driven ITD models.
- To examine the major technical challenges and limitations of applying XAI-based for ITD.
- To uncover existing research gaps and unresolved issues within the XAI-ITD area.
- To highlight key insights and outline future research directions for advancing interpretable, trustworthy ITD systems.

B. KEY CHALLENGES

This survey does not merely summarize previous reviews; rather, it emphasizes the analysis of technical studies in which XAI techniques have been empirically applied and evaluated in the context of ITD. By examining the results and limitations reported in these experimental works, we identify and highlight key challenges that persist in the development and deployment of XAI-based ITD systems. Despite increasing attention in both academia and industry, the XAI-ITD domain continues to face several critical challenges:

- **High false positive rates:** Many existing ITD systems rely on predefined behavioral indicators such as abnormal login times, unauthorized access to files, or atypical device usage patterns–to flag suspicious activity [11], [12]. While such rules can help identify certain anomalies, they often lack contextual nuance,

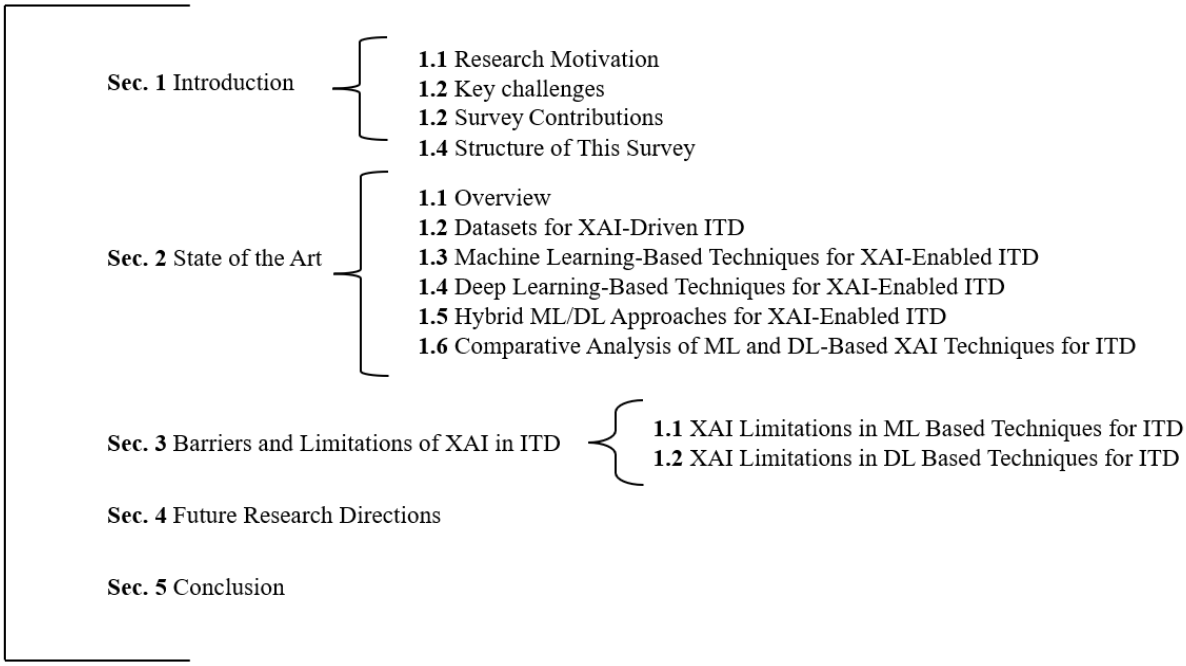


FIGURE 2. The structure of this survey.

resulting in a high volume of false positives and alert fatigue among analysts.

- **Contextual blindness:** Traditional detection models frequently ignore key contextual factors such as the semantic content of emails, website browsing behavior, and psychological indicators that could signal malicious intent [13]. Without incorporating this broader context, models risk missing subtle or covert insider behaviors.
- **Lack of interpretability:** A significant number of AI-based ITD systems, particularly those built using ML and DL, function as "closed boxes." Their decisions are often opaque, offering little to no insight into why a specific user or activity was flagged. This lack of transparency reduces trust and hinders effective decision-making by human analysts [13].

C. SURVEY CONTRIBUTIONS

This survey distinguishes itself by addressing a critical gap in the literature at the intersection of XAI and malicious ITD. While earlier surveys such as Arrieta et al. [14] provide conceptual taxonomies and foundational definitions of XAI, they lack domain specificity and do not consider the practical challenges of applying XAI in real-world cybersecurity contexts. Similarly, ITD-focused reviews like Salem et al. [15] and Sanzgiri et al. [16] primarily explore detection techniques without systematically examining the role of interpretability in supporting trust and operational transparency. In contrast, our work provides a focused, structured, and up-to-date review of XAI methods as applied to ITD, with the following contributions:

1) We present the rationale for integrating XAI into ITD systems and provide foundational background on widely adopted techniques such as SHAP and LIME.
2) We provide both a comprehensive summary and a concise overview of publicly available datasets suitable for applying XAI techniques in ITD.
3) We categorize ITD approaches into traditional ML, DL, and hybrid models, examining how XAI enhances model interpretability and supports analyst understanding.
4) We discuss various XAI approaches and highlight their advantages and limitations in the development of XAI-based ITD systems.
5) We justify the use of XAI for ITD by demonstrating its applicability across various real-world scenarios.
6) We outline the critical challenges and unresolved issues at the intersection of XAI and cybersecurity, and identify key insights and future research directions for advancing XAI applications in

This survey provides a comprehensive, gap-aware review that advances the current state of research by aligning XAI techniques with the specific demands, data modalities, and operational constraints of ITD.

D. STRUCTURE OF THIS SURVEY

As illustrated in Fig. 2, this survey is systematically structured to provide a comprehensive understanding of XAI techniques for ITD. The paper begins with Section I, which outlines the research motivation, key Challenges, contributions, and the overall structure of the survey. Section II presents a

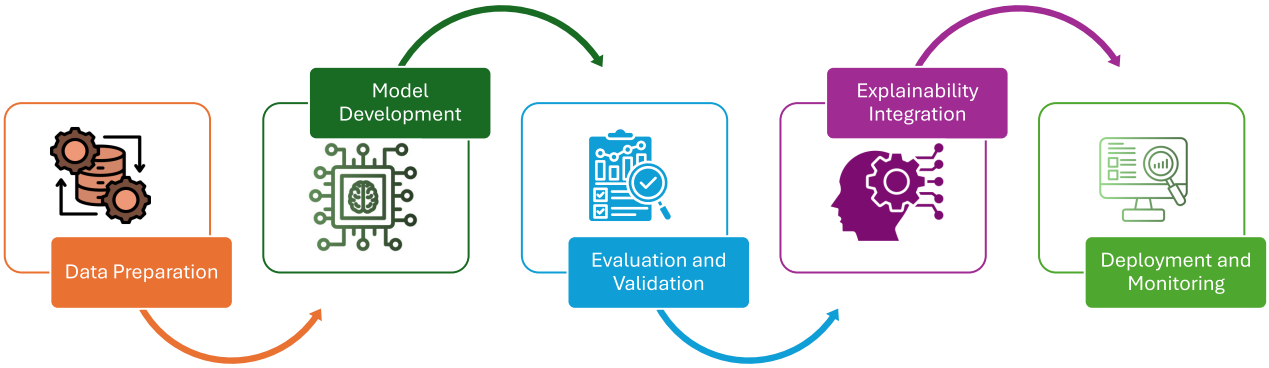


FIGURE 3. Overview of explainable artificial intelligence framework.

comprehensive review of existing XAI-enabled ITD techniques, covering ML, DL, and hybrid-based approaches. This section discusses the most widely used datasets, explains how they support explainable modeling, outlines common evaluation metrics (e.g., accuracy, precision, recall), and provides a comparative analysis of the reviewed techniques. Section III identifies and discusses the barriers and limitations encountered when applying ML and DL-based XAI methods in ITD contexts. Section IV explores potential future research directions that can advance the development and deployment of interpretable and robust threat detection systems. Finally, Section V concludes the survey by summarizing the major findings and emphasizing the importance of balancing accuracy, interpretability, and operational applicability in future XAI-enabled ITD solutions.

II. STATE OF THE ART

A. OVERVIEW

Insider threats remain among the most complex and pressing challenges in cybersecurity. These threats originate from individuals within an organization such as employees, contractors, or business partners. They abuse their authorized access to critical systems and sensitive data. This can potentially result in significant financial, reputational, or operational harm. Unlike external threats, which are typically initiated by adversaries exploiting vulnerabilities from outside the organization, insider threats are more difficult to detect due to their origin from users with legitimate credentials. These insiders may engage in malicious activities such as data exfiltration, intellectual property theft, or system sabotage, either intentionally or unintentionally. A central challenge in ITD is differentiating between benign and malicious user behavior, particularly when malicious intent is subtle and context-dependent. Mahbooba et al. [17] emphasize that detecting intent is inherently difficult due to the variability and ambiguity of insider behaviors. Koutsouvelis et al. [7] further highlight the importance of incorporating heterogeneous data sources–such as behavioral logs, communication patterns, and psychometric indicators–to enhance detection accuracy and reduce false positives.

Traditional detection methods, such as rule-based systems and basic anomaly detection techniques, suffer from several limitations. These include high false-positive rates, limited adaptability to evolving threat strategies, and a lack of transparency in the decision-making process. Such limitations reduce the trustworthiness and operational effectiveness of these systems. To address these gaps, recent research has increasingly adopted ML and DL approaches for ITD. Le and Zincir-Heywood [18] demonstrate that ensemble learning techniques improve model robustness, while Kumar et al. [19] explore the integration of psychometric data to create more comprehensive user behavior profiles.

Explainable AI has emerged as a critical component for enhancing the interpretability and reliability of ML/DL-based ITD systems. Unlike traditional closed-box models, XAI techniques provide human-understandable insights into how and why models make specific predictions, thereby increasing trust, accountability, and operational transparency–especially important in high-stakes domains like cybersecurity [20]. Tools such as Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) have been widely applied to demystify complex model decisions, offering both global and local interpretability. For example, SHAP has been used with models such as Random Forest and DNNs to provide consistent feature importance rankings across diverse datasets like DNSLog, UNSW-NB15, and CICIDS17 [21], [22], [23], while LIME has facilitated localized instance-level insights in models trained on IoT and behavioral datasets [24], [25]. To systematically address the challenges of integrating interpretability into ITD models, this study presents a generalized XAI-ITD pipeline. This pipeline synthesizes insights from existing frameworks. It builds on and extends conceptual workflows introduced in previous literature. One such example is the foundational XAI architecture outlined by Schlegel et al. [26]. The proposed pipeline is specifically tailored to the ITD domain. It incorporates five essential stages: (i) Multi-source data preparation–including behavioral logs, psychometric indicators, and communication records;(ii) Model development using traditional ML, DL, or hybrid methods; (iii) Evaluation based on predictive per-

formance and fidelity-oriented XAI metrics; (iv) Integrated interpretability using methods such as SHAP, LIME, RuleFit, and TreeLIME; (v) Deployment and continuous monitoring in evolving threat environments.

As illustrated in Fig. 3, the pipeline emphasizes not only performance but also traceability, explainability, and operational transparency. Key aspects of this framework include multimodal data fusion, real-time interpretability feedback, and compatibility with privacy-preserving strategies like federated learning [9], [25]. Unlike models that rely on static or one-time evaluations, this pipeline supports adaptive explainability. This makes it more suited for dynamic, real-world ITD scenarios. Overall, this generalized architecture offers a structured and practical foundation for applying XAI in cybersecurity. It is particularly useful where accountability and interpretability are as critical as detection accuracy.

Safeguarding cyber-physical systems like those in Smart Healthcare Systems (SHS) is challenging. This is due to the large volume of sensitive data exchanged between IoT devices, edge nodes, and cloud platforms. A blockchain-enabled XAI architecture offers a strong solution. It combines data integrity with transparent decision-making. Blockchain ensures tamper-evident logging. This means that any change to patient data can be detected. For example, when an IoT medical device sends sensitive health data to a cloud server, the blockchain validates the data. It checks the data's authenticity and integrity before allowing transmission. In parallel, XAI techniques offer transparency by explaining how risk scores or anomaly flags are generated, enabling security analysts and healthcare professionals to better understand and trust automated decisions. This dual-layered system not only improves threat detection and compliance but also strengthens trust across the healthcare ecosystem [19]. Sharma et al. [27] similarly examine the role of XAI in DL-based intrusion detection systems in dynamic environments like IoT networks. These systems benefit from XAI's ability to convert opaque model decisions into meaningful insights that support faster and more informed responses to threats.

ITD systems leverage a range of strategies, each utilizing distinct techniques and data modalities to identify behaviors indicative of insider risks. One foundational method, behavioral profiling, monitors user activity to establish baseline patterns and detect anomalies; this approach underpins User and Entity Behavior Analytics (UEBA) systems. However, UEBA systems frequently generate high false-positive rates due to the inherent variability in legitimate user behavior [28]. Rule-based and signature-based systems detect known threat scenarios, such as unauthorized file access or off-hours logins, but struggle to identify emerging or adaptive threat patterns without frequent rule updates [29]. Psychometric and personality-based methods–such as those based on the Five-Factor Model (OCEAN)–offer additional behavioral insights but also raise ethical and privacy concerns due to the sensitive nature of psychological profiling [30]. Natural Language Processing (NLP) has also been used to analyze unstructured data such as emails and chat transcripts, detecting shifts in sentiment or communication patterns that may suggest malicious intent. Advanced language models such as RoBERTa provide deeper contextual understanding, but face limitations in scalability and linguistic ambiguity [11], [31]. Graph-based methods model the relationships between users, devices, and systems as networks and have shown promise in uncovering coordinated insider activity. However, their effectiveness depends on significant computational resources and expert tuning to interpret abnormal patterns [29].

ML and DL models bring powerful pattern recognition capabilities to ITD. Unsupervised methods such as clustering and isolation forests have demonstrated success in identifying novel threats without labeled training data [44], [45]. However, these models are often viewed as opaque and difficult to interpret, which limits their practical use in high-stakes environments. Consequently, XAI has become a necessary addition to enhance their transparency and utility. Recent work by Kumar et al. [19] emphasizes the benefit of integrating psychometric profiles into ML pipelines, while Le and Zincir-Heywood [18] showcase the effectiveness of ensemble learning methods in boosting performance. In IoT contexts, Amiri Zarandi et al. [46] propose a privacy-preserving framework that combines federated learning with XAI to detect insider sabotage. Their model trains locally on distributed IoT devices to protect user data while employing LIME and SHAP to explain predictions. Their work also highlights the ongoing challenge of accessing realistic, large-scale insider threat datasets in IoT domains and the limitations of existing closed-box DL models in security applications.

The following sections expand on these insights by reviewing the latest research on XAI-ITD systems. We organize the discussion into focused parts to make the analysis clearer and more structured. These include: datasets commonly used for XAI-driven ITD, machine learning-based techniques, deep learning-based techniques, hybrid ML/DL approaches, and a comparative analysis of the strengths and limitations of these XAI methods. Each section is designed to highlight key trends in learning models, interpretability tools, and real-world applications.

B. DATASETS FOR XAI-DRIVEN ITD

ITD systems depend critically on the quality, diversity, and contextual depth of the datasets used for model development, training, and evaluation. In the context of Explainable AI, the relevance of these datasets extends beyond detection performance–they must also support interpretability, transparency, and actionable insights. As modern ITD approaches increasingly rely on ML, DL, or hybrid ML/DL models, the ability to integrate XAI tools such as SHAP, LIME, RuleFit, DeepLIFT, and Grad-CAM becomes essential. These tools require datasets with interpretable feature spaces, labeled behaviors, and temporal or behavioral granularity to generate

TABLE 1. Comparative overview of datasets for explainable AI in insider threat detection.

Paper Ref.	Dataset	Domain	Key Features	Insider Threat Use Case	Why/How Suitable for XAI-Based ITD
[32]–[34]	NSL-KDD	General Network	Traffic flows, attack types (DoS, R2L)	Anomaly detection, attack explanation	Interpretable, labeled data supports testing global and local explanations
[35], [36]	CICIDS2017	Realistic Network Traffic	Packet sizes, durations, connection patterns	Detection of known/unknown attacks	Rich flow-level metadata enables interpretable anomaly explanation
[9], [37]	UNSW-NB15	Hybrid Intrusion Detection	Modern labeled attacks, behavioral attributes	Advanced ML/DL explanation and training	Diverse attack types and feature sets suitable for model transparency
[24]	CICIoT2023	IoT Systems	Device communication, flow features	IoT insider threat modeling	Granular device data supports per-instance interpretability
[24]	IoTD20	IoT Environments	Interaction logs, traffic behaviors	Smart system insider tracking	Captures fine-grained behavior needed for behavior-aware XAI
[25]	ICU-IoT-Flock	IoT/Behavioral	Session/device interactions	High accuracy in device-level threat detection	Temporal session data enables interpretable event-based detection
[38]	Maldroid20	Mobile Security	Android permissions, APIs	Malware and mobile insider threats	Structured mobile feature space ideal for attribution-based XAI
[32]	5GNIDD	5G Network	Device/traffic interactions	5G-specific anomaly detection	Modern 5G structure enables explainable mobile access control
[39]	AWID	Wireless Networks	Authentication, access patterns	Wi-Fi intrusions and misuse	Logs access sequences that align with sequential explainability
[40]	Honeypot22	Deception-based	Attack traces from honeypots	Distinguishing internal vs. external threats	Mimics adversarial behavior ideal for XAI sensitivity testing
[41]–[43]	CERT r6.2	Organizational/ Behavioral	Logon, file, device, email, web logs	Scenario-based ITD	Multimodal behavioral logs and labeled scenarios ideal for temporal, contextual explanation

both global and instance-level explanations. In Fig. 4, we visualize the categorization of datasets discussed in this section, grouped according to their relevance to different ITD application domains.

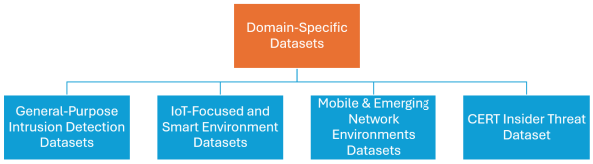


FIGURE 4. Domain-specific datasets for explainable insider threat detection.

General-purpose intrusion detection datasets such as NSL-KDD, CICIDS2017, and UNSW-NB15 have been widely used to develop both ML and DL-based ITD models. NSL-KDD provides a structured set of labeled attacks–Denial-of-Service (DoS), Remote-to-Local (R2L), User-to-Root (U2R), and probe attacks–alongside interpretable traffic features (e.g., protocol type, service, connection duration). Its clean labeling and relatively small feature space make it ideal for evaluating the output of explainability tools such as SHAP and LIME, which have been used in CNN and DNN contexts to identify which attributes most influence classification outcomes [23], [33], [36]. This interpretability makes it suitable for initial benchmarking of XAI techniques in ITD research. CICIDS2017 simulates real-world cyber attack scenarios (e.g., DDoS, brute-force) using realistic network traffic, with comprehensive flow-level features such as packet size, duration, and byte counts. Its rich contextual data enables meaningful attribution when SHAP or LIME is applied, supporting detailed post-hoc analysis of anomaly detection models 1. Studies like [35] show that models trained on CICIDS2017 and interpreted via XAI techniques can validate whether learned behaviors align with expected indicators, making it useful for validating model transparency in dynamic traffic environments. UNSW-NB15, developed to address the limitations of legacy datasets, incorporates modern network behaviors and diverse attack scenarios. It has been used with advanced classifiers such as XGBoost, MLP, and DT [56], and is particularly well-suited to XAI-based research due to its extensive feature space and updated labeling. Studies using SHAP and RuleFit have demonstrated how key behavioral features contributing to model predictions can be extracted and explained [9], [37], supporting fine-grained analysis of feature importance in complex detection pipelines.

TABLE 2. Key insights into ML and XAI integration for ITD.

Paper Ref.	AI Method(s)	Dataset(s) Used	Performance Metrics					XAI Techniques		
			Acc.	Prec.	Rec.	F1-Score	Other(s)	SHAP	LIME	Other(s)
[47]	ODT	Edge-IIoT	×	×	×	×	MCE=0.05	×	×	TreeLIME
[48]	FuzzyDT	N-BaIoT	×	0.99	0.99	0.99	×	×	×	FDTSemantics
[21]	RF	DNSLog	0.96	×	×	×	×	✓	✓	ANCHORS
[32]	RF	NSL-KDD, 5GNIDD	×	×	×	×	×	✓	✓	×
[34]	RF	NSL-KDD	0.98	0.87	0.99	0.98	×	✓	×	×
[49]	Meta Cluster	CICI-DS17	×	0.99	0.99	0.99	×	✓	✓	×
		Debrin	×	0.97	0.93	0.95	×	✓	✓	×
		CTI-TTPs	×	0.79	0.77	0.78	×	✓	✓	×
[50]	XGBoost	NSL-KDD	0.93	0.91	0.98	×	×	✓	×	×
[22]	XGBoost	DDoSAttackLog	0.97	0.98	0.96	0.97	×	✓	✓	×
[51]	KNN	LYCOSIDS2017	×	×	×	×	AUC=0.99	✓	×	×
[52]	SOMs	NSL-KDD	0.91	0.91	0.91	0.91	×	×	×	SOM Mappings
		CICI-DS17	0.80	0.77	0.82	0.80	×			
[53]	RFC	CICIDS18	0.98	0.99	0.98	0.99	×	✓	×	×
[54]	Ensemble (RF+DT+ SVM+ XGBoost)	EHMS2020	0.99	0.99	0.99	0.99	×	✓	×	×
[55]	SBPO+RF	NSL-KDD	0.99	0.99	0.99	0.99	×	✓	✓	×
[24]	Blending (RF+ GB+ DT)	CICIoT2023	0.99	0.98	0.99	0.99	×	×	✓	Counterfactual
		IoTD20	1	1	1	1	×			
[9]	FedIoT	UNSW-NB15	0.99	0.99	0.99	0.99	×	✓	×	RuleFit
[33]	DT, LR, RF	NSL-KDD	0.99	0.98	0.99	0.98	×	✓	✓	×
[37]	SKM-XGB	NSL-KDD	0.99	0.99	0.99	0.99	0.99	✓	×	×
		UNSW-NB15	0.99	0.98	0.97	0.98	×			

Datasets such as CICIoT2023, IoTD20, and ICU-IoT-Flock are tailored to smart and IoT environments where insider threats manifest as anomalies in device communication or access behavior. CICIoT2023, which captures device communication patterns and flow features, supports ensemble ML models and has been paired with LIME and counterfactual explanation techniques to enhance local interpretability and auditability of insider threat predictions [24]. Its detailed device-specific activity makes it ideal for scenario-based explainability studies. IoTD20 provides granular behavioral records from IoT systems, making it suitable for assessing global feature relevance with SHAP. While underexplored, the integration of LIME in future studies could further enhance per-instance explanation, which is critical in operational ITD settings where timely interpretation is needed [24]. ICU-IoT-Flock includes structured, session-level interaction logs that are ideal for both behavioral modeling and explainability. The dataset has demonstrated strong performance with RF, KNN, SVM, and ANN models (F1 scores up to 0.99) [25], and its structured temporal data allows both model-specific and model-agnostic explanation techniques (e.g., RuleFit, SHAP) to provide contextual insights about user-device interactions.

Mobile and next-generation network datasets such as Maldroid20, 5GNIDD, AWID, and Honeypot22 introduce new insider threat surfaces in increasingly decentralized infrastructures. Maldroid20, which includes Android malware samples, has been analyzed using DL models with

SHAP to identify critical API and permission usage patterns indicative of insider threats in mobile applications [38]. Its static analysis features allow SHAP to clearly attribute malicious behavior to specific app characteristics. 5GNIDD models traffic and device interactions in 5G environments, a growing area for insider threat research. With its protocol diversity and behavior-based labels, it is suitable for applying SHAP to uncover key predictors in high-volume, low-latency contexts [32]. AWID provides Wi-Fi access control and authentication logs, which are sequential in nature–making them suitable for deep models like CNN+LSTM and time-aware explainability methods. It enables analysis of session-based deviations and user behavior shifts that could indicate insider activity [39]. Honeypot22, which captures deceptive interaction data, supports the study of intent differentiation between insider and external threats. The application of SHAP and LIME in this dataset allows for interpretability of behavioral patterns that distinguish malicious insiders from false-flag attacks or red-team actors [40].

A notable highlight among ITD datasets is the CERT Insider Threat Dataset, developed by the Software Engineering Institute (SEI) at Carnegie Mellon University [57]. Unlike traditional intrusion detection datasets focused on external network attacks, CERT captures realistic insider threat scenarios in simulated enterprise environments. It includes rich behavioral logs such as logon/logoff activities, email usage, file access, HTTP browsing, and removable media interactions, with each user labeled as benign or malicious based on specific scenarios (e.g., data theft, sabotage, or policy violations) [41], [42]. The dataset's temporally structured and multimodal nature makes it well-suited for time-series modeling, behavioral profiling, and the application of explainable AI techniques such as SHAP, LIME, and DeepLIFT. These methods help identify which user actions contributed most to a model's decision, thereby supporting both global and local interpretability [43]. CERT remains one of the most authoritative and widely adopted benchmarks for building, validating, and interpreting explainable ITD models in research and operational settings.

C. MACHINE LEARNING-BASED TECHNIQUES FOR XAI-ENABLED ITD

ML approaches have become paramount in improving ITD by evaluating complex datasets and discovering subtle patterns that may indicate malicious activity. These approaches, ranging from standard supervised learning to more advanced unsupervised and hybrid models, have significantly enhanced the efficacy of ITD, as summarized in Table 2. Integrating XAI techniques into ML models ensures both accuracy and interpretability, enabling analysts to trust automated outcomes [53].

1) SUPERVISED LEARNING MODELS

Supervised learning models are widely utilized in ITD because they can classify behaviors as normal or suspicious based on labeled data. These models are trained on datasets that contain labels for each instance, allowing them to detect patterns associated with malicious behavior.

Decision Trees (DT) are widely regarded as effective supervised learning models for ITD due to their inherent simplicity and interpretability. These models operate by recursively partitioning data based on feature values like login times, file access frequencies, or device usage patterns, enabling transparent and traceable decision-making. This interpretability is particularly valuable in cybersecurity, where understanding the rationale behind predictions is crucial. For example, [47] demonstrated that an Optimal Decision Tree (ODT) applied to the Edge IIoT dataset achieved a low misclassification error (MCE) of 0.05. The integration of TreeLIME further enhanced its interpretability by providing localized, instance-level explanations. Similarly, in [48], a Fuzzy Decision Tree (FuzzyDT) achieved high precision, recall, and F1-scores of 0.99 on the N-BaIoT dataset. Its explainability was further strengthened by FDTSemantics, a framework that translates fuzzy rules into semantically meaningful insights. These studies highlight how Decision Tree-based models, when coupled with explainable AI techniques, can effectively balance performance with transparency. They do so by emphasizing critical behavioral indicators such as irregular login activity or anomalous file access.

Random Forest (RF) has emerged as a widely adopted supervised learning technique in ITD due to its scalability, robustness, and capacity to handle high-dimensional data. By constructing an ensemble of decision trees and using majority voting for classification, RF offers a balance between predictive performance and interpretability. In the study by Suryotrisongko et al. [21], RF was deployed on the DNSLog dataset. It achieved an accuracy of 0.96 in detecting malicious activities. To enhance model transparency, the authors integrated SHAP, LIME, and ANCHORS. These three post-hoc explainability methods helped clarify how feature contributions influenced classification outcomes. A similar application was presented in [32], where RF was trained on the NSL-KDD and 5GNIDD datasets. The model was used to detect threats in 5G network environments. SHAP and LIME were again employed to interpret predictions. This allowed analysts to identify the most influential variables within complex, evolving traffic patterns. This use case demonstrated RF's flexibility in addressing modern cybersecurity challenges posed by next-generation network infrastructures. Furthermore, in [34], RF achieved outstanding performance in ITD. SHAP was used to dissect the model's predictions. It offered analysts a transparent view of how specific behavioral indicators–such as unusual login times or irregular file access–contributed to threat classification.

The MetaCluster framework was evaluated by Ge et al. [49] across multiple datasets to assess its robustness and generalizability in detecting insider threats. In the first scenario, the model was tested on the CICIDS17 dataset, where it achieved near-perfect results with accuracy, precision,

recall, and F1-score all reaching 0.99. These metrics indicate the model's strong capability in identifying anomalous network behaviors with high reliability. When evaluated on the Debrin dataset, performance remained strong but showed slight degradation, with accuracy dropping to 0.97, precision to 0.93, and recall to 0.95. However, on the CTI-TTPs dataset, the MetaCluster method experienced a notable decline in performance–achieving 0.79 accuracy, 0.77 precision, and 0.78 recall. These results suggest that while the model performs well in structured and balanced settings, its effectiveness may diminish when faced with more heterogeneous or challenging data distributions. The study integrated SHAP and LIME. These tools provided transparency into the model's decision logic by highlighting the most influential features in each classification instance. By combining strong predictive performance with model explainability, MetaCluster demonstrated a balanced approach to ITD–capable of adapting across datasets while maintaining analyst trust through transparent insights.

Extreme Gradient Boosting (XGBoost) is widely used in cybersecurity. It is known for high accuracy, robustness, and the ability to handle large and complex datasets. Its ensemble-based approach makes it suitable for detecting insider threats and other malicious activities. These threats often involve subtle patterns in high-dimensional data. In a study by Barnard et al. [50], XGBoost was tested on the NSL-KDD dataset. The model achieved 0.93 accuracy, 0.91 precision, and 0.98 recall. These results show that the model can detect abnormal activities while keeping the false negative rate low. This is important to ensure that real threats are not missed. SHAP was used to explain how each feature affected the model's output. Another study by Jiyad et al. [22] applied XGBoost to the DDoSAttackLog dataset. The model achieved 0.97 accuracy, 0.98 precision, 0.96 recall, and a 0.97 F1-score. These metrics show that XGBoost performed well in identifying DDoS attacks while minimizing errors. SHAP was also used in this study.

The k-Nearest Neighbors (KNN) algorithm is a simple and effective method for ITD. It works by using similarity between data points to classify behavior. In the study by [51], KNN was used on the LYCOS-IDS2017 dataset. The model achieved an AUC score of 0.99, demonstrating KNN's strong ability to separate normal and malicious behavior. The study used SHAP to explain the model's decisions by showing how feature similarity influenced the classification results. This made the model's behavior easier to understand. The results show that KNN, when combined with XAI tools like SHAP, can detect insider threats while keeping the system transparent.

2) UNSUPERVISED LEARNING MODELS

Unsupervised learning methods are essential for identifying risks in scenarios where labeled data is limited or unavailable. These models are highly effective in detecting underlying patterns and anomalous behavior, which is particularly valuable in ITD due to the covert nature of many malicious actions. Unsupervised models can be integrated with XAI techniques not only to detect potential threats but also to provide explainable insights into their decision-making processes. In the study [52], the authors describe the application of Self-Organizing Maps (SOMs) for ITD. SOMs, an unsupervised learning algorithm used for clustering and dimensionality reduction, can effectively identify patterns and outliers in large datasets. The SOM model was evaluated using the NSL-KDD dataset for network intrusion detection, achieving favorable results: an accuracy of 0.91, with precision, recall, and F1 score values also equal to 0.91. This demonstrates the model's capability in identifying insider threats. In another instance, the SOM model was tested on the CICIDS17 dataset, yielding slightly lower performance metrics: an accuracy of 0.80, precision of 0.77, recall of 0.82, and an F1 score of 0.80. In this evaluation, SOM mappings were employed to explain the underlying structure of the dataset. Although the study did not incorporate advanced explainability techniques such as SHAP or LIME–which could further enhance interpretability–SOMs still proved useful for clustering and mapping data into meaningful groupings. Their capacity to uncover subtle anomalies makes them a valuable tool for ITD, even in the absence of modern XAI tools [52].

3) HYBRID ML MODELS

Incorporating different algorithms has demonstrated outstanding success in identifying insider risks. By integrating the capabilities of diverse methodologies, these models improve accuracy, robustness, and adaptability, key characteristics needed to address the complex and dynamic nature of insider threat behaviors. The paper by [53] proposed a novel IDS architecture that merges XAI with Random Forest Classifiers (RFC) to improve both performance and interpretability. The study compared a baseline RFC model with an enhanced IDS that utilized SHAP to increase transparency and trustworthiness. On the CICIDS18 dataset, the baseline RFC model achieved excellent results: 0.98 accuracy, 0.99 precision, 0.98 recall, and 0.99 F1-score.

Another study established a combination of multiple distinct models for ITD. The work in [54] introduced an ensemble model consisting of RF, DT, Support Vector Machines (SVM), and XGBoost. This ensemble proved highly effective on the EHMS2020 dataset. By leveraging the strengths of each algorithm, the ensemble effectively addressed various dimensions of network traffic and user activity, offering a comprehensive solution for detecting insider threats.

Similarly, [55] implemented the SBPO+RF model, attaining 0.99 in all key performance metrics. This approach not only improved model performance but also incorporated SHAP and LIME for explainability. These techniques provided clear and interpretable insights into the model's decision-making, ensuring transparency and enabling timely, informed responses to potential risks.

In [24], A combined method utilizing RF, Gradient Boosting (GB), and DT achieved optimal results on the IoTD20 dataset, with accuracy, precision, recall, and F1-score all equal to 1. This hybrid framework effectively captures the diverse and complex behaviors typical of IoT environments. The FedIoT model, built on a federated learning architecture, also produced excellent results on the UNSW-NB15 dataset, achieving 0.99 across all major evaluation metrics [9]. To address the challenge of interpretability, the study employed RuleFit for feature selection, enabling the model to justify its decisions while detecting even small volumes of malicious activity–without compromising data privacy.

Additionally, [33] developed a hybrid model combining DT, RF, and Logistic Regression (LR), achieving 0.99 accuracy and 0.98 precision on the NSL-KDD dataset. The integration of SHAP and LIME further improved the interpretability of model predictions. This ensemble approach is effective because it captures both linear and non-linear relationships in the data–an essential feature for accurate ITD.

The SKM-XGB model introduced in [37] also achieved consistently high performance, with approximately 0.99 reported across all metrics in two separate test scenarios. The proposed solution, which integrates the predictive power of XGBoost with the structural learning capabilities of SKM-based approaches, proved to be highly effective in detecting insider threats in large-scale data environments. These findings highlight the advantages of combining multiple algorithms to enhance detection capabilities and reduce false alarms.

Finally, [24] also proposed a hybrid model evaluated on the CICIoT2023 dataset, attaining near-perfect performance with 0.99 accuracy and F1-score. This model uniquely incorporated counterfactual reasoning a novel element not present in previous frameworks. It provided a more intuitive and precise analysis of the decision-making process by illustrating how small variations in input features could influence model predictions.

4) INTEGRATION OF EXPLAINABILITY TOOLS (SHAP, LIME AND OTHERS) WITH ML LEARNING MODELS

With the growing complexity of AI models, there is an increasing need to ensure explainability during the decision-making process, particularly in sensitive domains such as ITD. This section examines the explainable AI techniques presented in Table 2.

SHAP is one of the most widely used techniques for explaining ML models, providing both global and local explanations. In the area of ITD, SHAP has proven useful for enhancing the interpretability of ML models, particularly in identifying how specific features influence model outputs. As reported in [50], the NSL-KDD dataset was analyzed using XGBoost, and model explanations were generated using SHAP. The study demonstrated how SHAP provided insights into the importance of features such as login frequency and data access patterns. These insights helped determine how particular behaviors were associated with normal or malicious activity. In this context, SHAP played a significant role in increasing confidence in the XGBoost model, especially when dealing with large and complex datasets where other interpretability methods may fall short. SHAP improved the interpretability and trustworthiness of XGBoost in ITD by clarifying the contribution of individual features. Similarly, in [51], KNN was employed for ITD, and SHAP was incorporated to interpret the model's outputs. The use of SHAP made it possible to identify which features most significantly influenced classification–such as user access times or network activities. In terms of interpretability, although KNN is generally considered a less complex algorithm, SHAP provided the necessary level of transparency to detect insider threats based on user behavior. This demonstrates that SHAP can support interpretability across both simple and complex models in the context of insider threat analysis. In [9], the UNSW-NB15 dataset was used with a model called FedIoT, a federated learning-based approach, and SHAP was applied for interpretability. Federated learning, which involves training models on decentralized data sources, presents unique challenges for interpretability due to its distributed nature. However, SHAP enabled the study to reveal how features from various sources contributed to overall model predictions. This capability is essential for making federated models trustworthy and interpretable–particularly in scenarios that require detecting insider threats across organizational boundaries.

LIME is one of the most widely used techniques for explaining the predictions of ML models by focusing on local, instance-based explanations. While methods such as SHAP provide a broader approach to interpreting the relative importance of features in a model, LIME offers an instance-specific perspective on how features influence outcomes for particular cases. This is especially important in ITD, where identifying even minor deviations from normal behavior in user interactions or network traffic is critical. In [24], the authors employed a blending model–comprising several ML techniques–on the CICIoT2023 dataset. LIME was applied to generate local explanations for each prediction, allowing analysts to observe how changes in specific features (e.g., device usage or network traffic) would affect the model's decisions. Although SHAP was not used in this case, LIME proved effective in explaining the decision-making process at the individual case level, particularly within IoT environments, where the behavior of individual devices is crucial for identifying insider threats. LIME enhanced the interpretability of the model and enabled security specialists to investigate specific instances of suspicious activity, understanding the rationale behind why they were flagged.

By combining SHAP's global perspective with LIME's instance-level explanations, the model became interpretable at scale and actionable on a case-by-case basis. This significantly improved its ability to detect insider threats. A RF model was applied to the DNSLog dataset in [21]. SHAP provided a global interpretation of feature importance. It highlighted the influence of factors such as network traffic

TABLE 3. Summary of explainability techniques applied to ML models for ITD.

Paper Ref.	XAI Technique(s)	Strengths	Limitations
[50]	SHAP	- Explains XGBoost on NSL-KDD - Identifies key features like login frequency and access patterns - Enhances trust in complex datasets	- Limited to feature importance - Offers only global explanations
[51]	SHAP	- Interprets KNN for ITD - Highlights critical features such as access time	- Less effective for non-tabular data
[9]	SHAP, RuleFit	- SHAP interprets federated FedIoT model globally - RuleFit generates human-readable decision rules	- Increased complexity due to decentralized setup (SHAP, RuleFit)
[24]	LIME, Counterfactual	- LIME: Local explanations for predictions - Counterfactuals show decision boundary thresholds	-
[21]	SHAP, LIME	- SHAP: Global feature impact via SHAP on RF - LIME: Instance-level insights via LIME	- Combined use increases computational overhead (SHAP, LIME)
[32]	SHAP, LIME	- SHAP identifies key RF features like flow and connection duration - LIME provides local interpretability for predictions	- Higher runtime complexity in 5G environments (SHAP, LIME)
[33]	SHAP, LIME	- SHAP explains feature importance across ensemble - LIME enables evaluation of flagged activities	- Limited generalizability across datasets (SHAP, LIME)
[22]	SHAP, LIME	- SHAP shows influence of traffic features - LIME explains specific DDoS predictions	- High computational cost (SHAP, LIME)
[47]	TreeLIME	- Tailored for Decision Trees (TreeLIME) - Simplifies ODT interpretation on Edge-IIoT (TreeLIME)	-

load and access mode on the model's outcomes. Conversely, LIME offered local interpretations. This allowed analysts to understand why specific network interactions were classified as malicious or benign. Similarly, in [32], the authors investigated ITD in 5G networks using the 5GNIDD dataset and an RF model. SHAP identified key features that contributed to threat detection, such as traffic flow, connection duration, and user engagement. LIME provided complementary local explanations. These helped analysts understand the rationale behind individual predictions. This combined methodology improved both the interpretability and accuracy of the model. It was especially effective in detecting anomalies in complex 5G environments. In another study, [33] used an ensemble model comprising DT, LR, and RF algorithms on the NSL-KDD dataset for intrusion detection. SHAP was used for global interpretability. It identified important features such as packet size, connection type, and session duration. LIME helped examine specific predictions, allowing analysts to investigate and validate flagged activity. Finally, [22] applied XGBoost to the DDoSAttackLog dataset to detect DDoS attacks. SHAP revealed that traffic-related features–such as packet arrival rates and connection frequencies–were critical to the model's classification logic. LIME complemented this by explaining individual predictions. It clarified why certain traffic instances were classified as DDoS attacks. Together, SHAP and LIME made the model both interpretable and effective. They were especially useful in identifying anomalies within DDoS contexts.

In addition to SHAP and LIME, several other XAI techniques have been integrated into ML models to enhance their interpretability, particularly in ITD. These alternative methods provide valuable insights into model behavior, further improving transparency and helping analysts better understand the reasoning behind model predictions. In [24], counterfactual explanations were used alongside a blending model applied to the CICIoT2023 dataset. While SHAP was not employed in this case, counterfactual explanations offered a unique approach to interpreting the model's predictions by illustrating how slight changes in input features could lead to different outcomes. This technique enabled analysts to understand what modifications would be necessary to alter the model's classification of a given instance, providing a more intuitive means of exploring the model's decision boundaries. Counterfactuals are particularly useful in cybersecurity applications, where understanding the threshold at which non-malicious behavior becomes malicious is crucial. In [47], TreeLIME was introduced as a method specifically designed for explaining the predictions of decision tree-based models, such as the Optimal Decision Tree (ODT) applied to the Edge-IIoT dataset. TreeLIME, a variant of LIME tailored for decision trees, provides local interpretability by approximating the decision boundary of

TABLE 4. Key insights into DL and XAI integration for ITD.

Paper Ref.	AI Method(s)	Dataset(s) Used	Performance Metrics					XAI Techniques		
			Acc.	Prec.	Rec.	F1-Score	Other(s)	SHAP	LIME	Other(s)
[58]	DNN	CICI-DS17, IoT Data	0.99	0.99	0.98	0.99	×	✓	x	x
[59]	DNN	N-BaIoT	0.99	0.99	0.99	0.98	×	x	x	DeepLift
[23]	DNN	NSL-KDD UNSW-NB15	0.88 0.99	0.96 ×	0.88 ×	0.88 ×	× TPR=0.99	✓	✓	RuleFit
[60]	DNN	UNSW-NB15	x	x	x	x	×	✓	x	RuleFit
[61]	DNN	logicGAN	x	x	x	x	×	x	x	Explanation Matrix
[36]	CNN	NSL-KDD CICI-DS17 SIMARGL2021	0.73 0.98 0.99	0.65 0.82 0.99	0.49 0.65 0.99	0.48 0.66 0.99	AUC=0.65 AUC=0.65 AUC=0.99	✓	✓	x
[62]	CNN	Public pCap	0.97	x	x	x	×	×	×	Genetic Algo
[63]	CNN	PSIB, ISH, Spa-mArchive, TREC	0.97	0.99	0.96	0.97	×	✓	✓	x
[35]	Auto-Encoder	CICI-DS17	0.92	0.94	0.92	x	×	x	x	DICE
[64]	Auto-Encoder	CICI-DS17	0.90	0.93	0.90	0.91	×	x	x	kernelSHAP
[65]	MLP	ADFA-LD	0.94	0.98	x	0.94	Sensitivity =0.894 Specificity =0.98	✓	✓	x
[66]	FeedFwd-NN	NIDKaggle	0.94	0.95	x	0.94	×	✓	✓	x
[67]	FeedFwd-NN	KD99	x	x	x	x	×	x	x	TCAV
[68]	GNN	OT Data	x	x	x	x	AUC=0.84 TP=1 FP-Reduc.=0.93	x	x	DL-Learner, GNNExplainer
[69]	VG19, Xception, Vit+CNN	Custom	0.98	x	x	0.98	×	x	x	IG, Grad-CAM
[70]	BERT	MM-Covid19	0.93	0.93	0.93	0.92	×	✓	x	x
[38]	XENIA (FGSM+ DNN+TL)	CICI-DS17 Maldroid20	x x	x x	x x	0.95 0.89	x x	✓	x	x
[39]	CNN+LSTM	NSL-KDD, CICI-DS17, AWID	0.99	0.57	0.31	0.41	×	✓	x	x

TABLE 4. *(Continued.)* **Key insights into DL and XAI integration for ITD.**

[40]	BO-GRU-LTSM + EFFS	NSL-KDD CICI-DS17 Honeypot22	0.97 0.96 0.98	0.93 0.94 0.95	0.94 0.96 0.96	0.95 0.80 0.97	x x x	✓	x	x
[71]	CNN+CIL	IoT23	x	x	x	x	F1-Drop=0.06 PAUC-Drop=0.01 AUC-Drop=0.02	✓	x	x
[72]	CNNs-GRU-FCN	CIC-Darknet-2020	0.99	0.99	x	0.99	×	✓	✓	x
[73]	X-HDNN	WUSTL-IIoT-21	0.93	0.78	0.77	0.78	×	✓	x	x
[74]	Compact-ANN	IoT-CyberSec	0.85	x	x	x	×	✓	✓	x
[75]	FL-DRL	O-RAN	x	x	x	x	×	x	x	ORAN Maps
[76]	EnsembleDL	NSL-KDD CICI-DS17 UNSW-NB15 MalDroid20	0.99 0.98 0.69 0.84	x x x x	x x x x	0.99 0.99 0.68 0.84	x x x x	x	x	Dalex
[77]	SALSTM+BC	N-BaIoT DTs	0.99 0.96	0.96 0.88	0.96 0.85	x x	x x	✓	x	x
[78]	FNN+BC	LVDAndro	0.96	0.94	0.99	0.96	×	x	✓	x

the tree with a simple, interpretable model. This method was used to explain how specific device behaviors or user actions in an IoT environment were classified as anomalous, enabling security analysts to gain a clear understanding of why particular decisions were made. The use of TreeLIME underscores the importance of applying specialized interpretability techniques to specific types of models, ensuring the explanation process remains effective even for more complex tree-based architectures. In [9], RuleFit was applied to the FedIoT model, which utilized federated learning to detect insider threats in decentralized IoT environments. While SHAP was employed for global interpretability, RuleFit was integrated to improve model transparency by generating human-readable rules that explained how various features influenced predictions. RuleFit identifies the most important features and combines them into simple decision rules, providing security analysts with a straightforward understanding of the model's reasoning. This approach is particularly advantageous in environments where interpretability is essential but the underlying model complexity could otherwise hinder comprehension.

D. DEEP LEARNING-BASED TECHNIQUES FOR XAI-ENABLED ITD

The integration of deep learning models with XAI is gradually gaining significance in mitigating the impact of insider threats. The application of DL in threat prediction, combined with the integration of XAI to create robust and transparent systems, enables researchers to develop interpretable frameworks for managing insider threats in complex environments. Table 4 presents the DL models used in this context.

1) NEURAL NETWORKS (NNS) MODELS

Neural networks, particularly Deep Neural Networks (DNNs), are widely recognized for their capability to handle complex and high-dimensional data, making them a cornerstone in ITD. Their applications span various domains,

including network intrusion detection, IoT security, and industrial anomaly detection. In the following, we detail studies that have employed DNNs, highlighting their performance and the integration of explainable AI techniques to address interpretability challenges.

DNNs have demonstrated exceptional performance across a range of datasets. In [58], a DNN applied to IoT data achieved outstanding metrics, including an accuracy of 0.99, precision of 0.99, recall of 0.98, and an F1 score of 0.99. This highlights the model's ability to effectively distinguish between normal and malicious behaviors in IoT environments. Similarly, [59] applied a DNN to the N-BaIoT dataset and achieved near-perfect performance, with precision, recall, and F1-score all approximately 0.99, confirming the model's capability in identifying malicious activities from IoT environments.

In [23] and [59], DNNs were effectively utilized to detect malicious behavior within network traffic, demonstrating strong performance in classifying complex patterns. Specifically, the model in [23] achieved accuracy values of 0.88 and 0.99 on the UNSW-NB15 and CICIDS17 datasets, respectively, along with a precision of 0.96 and a true positive rate (TPR) of 0.99–highlighting its robustness in real-world intrusion detection tasks.

In a related study, [60] explored the use of DNNs with the UNSW-NB15 dataset while integrating post-hoc interpretability techniques such as SHAP and RuleFit. Although exact performance metrics were not disclosed, the focus was placed on enhancing the model's transparency, allowing analysts to understand how decisions were made. Collectively, these studies underscore both the effectiveness and the explainability challenges associated with DNN-based models in cybersecurity. The use of XAI tools such as SHAP, LIME, and RuleFit provided valuable insights into model behavior, supporting more accountable and interpretable AI-driven security systems.

In [61], a DNN was employed on the LogicGAN dataset, accompanied by innovative explanation methods such as explanation matrices, which illustrate the contribution of specific features. This approach improved transparency, particularly in scenarios where understanding the influence of individual features is critical.

2) CONVOLUTIONAL NEURAL NETWORKS (CNNS) MODELS

Convolutional Neural Networks (CNNs) are highly effective for extracting spatial and temporal features, making them particularly suitable for ITD. Using convolutional layers, CNNs automatically identify and learn patterns in structured data such as network logs, packet captures, and user activity records. Their ability to process both spatial and sequential data enables them to detect anomalies that may indicate insider threats. In the following, we explore how CNNs have been applied in various studies for ITD, focusing on their performance and the integration of explainable AI tools to improve interpretability.

In [36], a CNN was applied to the CICIDS17 dataset, achieving an accuracy of 0.98, with corresponding precision, recall, and F1-scores of 0.82, 0.65, and 0.66 respectively. These results suggest that, when properly optimized through preprocessing and hyperparameter tuning, CNNs are capable of capturing intricate relationships in high-dimensional datasets. To address the opacity typically associated with deep learning models, the study also utilized SHAP and LIME to explain prediction outcomes, highlighting the importance of individual feature contributions in identifying malicious behavior.

In a parallel experiment conducted on the SIMARGL2021 dataset, CNNs demonstrated even stronger performance. As reported in [36], the model achieved near-perfect classification results, with precision, recall, F1-score, and AUC values all reaching 0.99. Once again, the integration of SHAP and LIME contributed to a deeper understanding of how the model arrived at its decisions, particularly by identifying influential communication patterns that played a central role in threat detection. These findings reaffirm CNNs' capacity to manage highly complex detection tasks while maintaining a level of interpretability that is essential for operational trust.

In addition, [62] explored a hybrid approach that combined CNNs with a genetic algorithm for feature optimization. This method was applied to public pCap datasets and achieved an accuracy of 0.97. Although the study did not report detailed precision or recall values, the use of evolutionary algorithms for feature selection significantly enhanced the model's ability to detect nuanced patterns, further demonstrating the adaptability of CNNs to complex cybersecurity environments.

Furthermore, the study in [63] extended the application of CNNs across diverse datasets, including PSIB, ISH, SpamArchive, and TREC. The model yielded strong results, achieving 0.97 accuracy, 0.99 precision, 0.96 recall, and an F1-score of 0.97. In line with other studies, SHAP and LIME were employed to interpret the results, offering visibility into the underlying decision-making process. This interpretability is essential, especially in high-stakes cybersecurity domains where model transparency must accompany predictive accuracy to ensure accountability and trust.

3) ADDITIONAL DEEP LEARNING BASED APPROACHES

Beyond the widely studied DNNs and CNNs, various DL architectures have been employed in cybersecurity for detecting insider threats and abnormal behaviors in complex environments. These models–including Auto-Encoders, Multi-Layer Perceptrons (MLPs), Feedforward Neural Networks (FeedFwd-NNs), Graph Neural Networks (GNNs), hybrid convolutional models, and BERT-based transformers–offer distinct advantages in handling large-scale, heterogeneous datasets. Moreover, the integration of Explainable AI tools with these models enables greater transparency and trust in decision-making.

Auto-Encoders have been widely adopted in cybersecurity applications due to their capability to learn compressed

representations and detect anomalies based on reconstruction errors. In [35], an Auto-Encoder applied to the CICIDS17 dataset achieved an accuracy of 0.92, precision of 0.94, and recall of 0.92. To support interpretability, the DICE framework was used to visualize reconstruction discrepancies and highlight anomalies. Similarly, the study in [64] implemented an Auto-Encoder on the same dataset, reaching an accuracy of 0.90, precision of 0.93, recall of 0.90, and an F1-score of 0.91. KernelSHAP was integrated into the model to explain feature contributions, providing interpretable diagnostics that helped security analysts understand the detection rationale.

Multi-Layer Perceptrons (MLPs) have proven effective in supervised classification tasks, particularly in detecting system-level anomalies. In [65], an MLP trained on the ADFA-LD dataset reached an accuracy of 0.94, precision of 0.98, and an F1-score of 0.94. The model demonstrated strong sensitivity (0.894) and specificity (0.98), minimizing false negatives while preserving high accuracy. SHAP and LIME were employed to generate local and global explanations, revealing the importance of input features such as specific system calls and file access patterns.

Feedforward Neural Networks (FeedFwd-NNs) offer a lightweight yet powerful architecture for pattern recognition in cybersecurity data. In [66], a FeedFwd-NN applied to the NIDKaggle dataset achieved an accuracy of 0.94, precision of 0.95, and an F1-score of 0.94. The integration of SHAP and LIME helped clarify how attributes like network flow characteristics contributed to classification outcomes. Another approach, described in [67], applied a FeedFwd-NN to the KD99 dataset using Testing with Concept Activation Vectors (TCAV) to provide high-level, human-interpretable concepts–facilitating transparency for non-technical stakeholders.

Graph Neural Networks (GNNs) are especially useful for modeling relationships in graph-structured data, such as communication logs or user access graphs. In [68], a GNN-based model was applied to Operational Technology (OT) datasets. The use of DL-Learner and GNNExplainer enabled the system to maintain a true positive rate (TPR) of 1 and reduce false positives with a reduction ratio of 0.93. These interpretability tools allowed researchers to understand how the GNN mapped user relationships to detect insider threats, offering a clear and transparent analytical framework.

In [69], a hybrid DL model combining VG19 and Xception architectures was introduced. The system achieved an accuracy and F1-score of 0.98 on a customized dataset. This approach capitalized on the strengths of both convolutional models to classify nuanced patterns. Integrated Gradients and Grad-CAM were employed to visualize the relevance of input features such as traffic segments or attack patterns, allowing security analysts to comprehend and verify predictions with confidence.

Originally developed for natural language processing, BERT (Bidirectional Encoder Representations from Transformers) has also been adapted for cybersecurity scenarios involving text-based data. In [70], BERT was applied to the MM-Covid19 dataset to analyze insider threats within text communications, achieving accuracy, precision, and recall scores of 0.93, along with an F1-score of 0.92. SHAP was used to interpret the model's predictions, illustrating which linguistic patterns or semantic cues contributed to risk classification. This adaptation of BERT demonstrates the model's utility in environments rich with textual logs and communication data.

4) HYBRID DEEP LEARNING-BASED APPROACHES

Integrating multiple DL models enables cybersecurity researchers to harness the strengths of different architectures, leading to enhanced accuracy, adaptability, and robustness–particularly in complex and evolving threat landscapes. This approach is especially useful in ITD, where diverse data modalities and intricate behavior patterns often exceed the capacity of single-model systems.

In [71], a hybrid CNN+CIL (Contextual Interactive Learning) model was applied to the IoT23 dataset, focusing on interpretability under feature perturbation. The study revealed a slight performance drop–0.06 in F1-score, 0.01 in PAUC, and 0.02 in AUC–when key features were altered, indicating the model's resilience to data drift. SHAP was used to identify the contextual dependencies within IoT traffic, highlighting how feature interactions influenced decision-making. This method proves effective for dynamic environments, where the relevance of input features evolves over time.

Another hybrid approach was proposed in [39], where CNN and LSTM were combined to extract spatial and temporal patterns, respectively, from the AWID dataset. While the model achieved a high accuracy of 0.99, other metrics were comparatively lower (precision = 0.57, recall = 0.31, F1-score = 0.41). This disparity underscores the challenge of balancing detection precision with generalization in real-world datasets. Nevertheless, SHAP was employed to enhance interpretability, offering a clearer understanding of which temporal-spatial features influenced predictions.

A more sophisticated ensemble architecture was presented in [40], where a Bidirectional Optimized GRU-LSTM model was paired with Enhanced Feature Selection Strategies (EFFS). The model yielded robust results across multiple datasets. On the Honeypot22 dataset, it reached an accuracy of 0.98 and an F1-score of 0.97; on NSL-KDD, it achieved 0.97 accuracy and a 0.95 F1-score; and on CICIDS17, it obtained 0.96 accuracy and an F1-score of 0.80. SHAP, combined with feature selection, enabled the identification of high-impact features contributing to malicious behavior–thus supporting both performance and interpretability.

In [38], the XENIA model was introduced, combining the Fast Gradient Sign Method (FGSM) with DNN and transfer learning techniques. Applied to the MalDroid20 dataset, the model achieved an F1-score of 0.89 and 0.95 on CICIDS17. FGSM improved resilience against adversarial examples, while SHAP clarified how specific features contributed to classification decisions. The integration of transfer learning

allowed the model to adapt effectively to novel environments with minimal retraining.

A multi-component architecture was explored in [72], where CNNs, Gated Recurrent Units (GRUs), and Fully Connected Networks (FCNs) were combined and applied to the CIC-Darknet-2020 dataset. This model achieved 0.99 across accuracy, precision, and F1-score. The hybrid design allowed simultaneous learning from spatial, sequential, and dense features. SHAP and LIME were both incorporated to provide local and global explanations of the model's behavior–making it suitable for security analysts investigating anomalies in encrypted or obfuscated traffic.

The X-HDNN (eXtreme Hierarchical Deep Neural Network) introduced in [73] focused on industrial IoT environments using the WUSTL-IIoT-21 dataset. The model reached an accuracy of 0.93, precision of 0.78, and an F1-score of 0.78. Despite the absence of post-hoc explainability tools such as SHAP or LIME, the model demonstrated practical effectiveness in handling multi-layered IIoT structures–suggesting a foundation for future interpretable designs.

A lightweight approach was proposed in [74], where the Compact-ANN model was evaluated on the IoT-CyberSec dataset. With an accuracy of 0.85, the model prioritized computational efficiency over complexity. SHAP and LIME were applied to explain the model's predictions, enabling analysts to understand which features–despite being minimal–were most influential. This trade-off highlights the importance of model efficiency in edge-device scenarios without sacrificing interpretability.

In [75], the FL-DRL (Federated Learning Deep Reinforcement Learning) hybrid model was applied to the O-RAN dataset. Although detailed performance metrics such as accuracy, precision, and recall were not reported, FL-DRL addresses the challenge of decentralized learning in mobile network environments. It employs federated learning to train the model across distributed devices without centralizing data, thereby preserving privacy. The reinforcement learning component allows the model to optimize actions based on reward signals, making it well-suited for real-time intrusion detection in network traffic.

Ensemble DL integrates multiple DL architectures to leverage their combined strengths, yielding more robust predictions. In [76], ensemble DL models were applied to datasets including UNSW-NB15, MalDroid20, NSL-KDD, and CICIDS17. Using the UNSW-NB15 dataset, the model achieved an F1-score of 0.68; for MalDroid20, the F1-score reached 0.84. On the NSL-KDD and CICIDS17 datasets, the models achieved accuracies and F1-scores of up to 0.99. The use of Dalex as an XAI tool enabled extensive feature importance explanations, allowing analysts to trust model outputs and identify key indicators of insider threats.

In [77], the SALSTM+BC (Self-Attentive LSTM with Boosted Classifiers) hybrid model was applied to the N-BaIoT dataset and a custom DTs dataset. On N-BaIoT, the model achieved an accuracy of 0.99 and precision and recall of 0.96. On the DTs dataset, the model achieved slightly lower metrics, with an accuracy of 0.96 and precision of 0.88. The inclusion of self-attention mechanisms allowed the model to capture complex dependencies in IoT data, while SHAP provided interpretability by identifying the most critical features.

The FNN+BC (Feedforward Neural Network with Boosted Classifiers) hybrid model presented in [78] was applied to the LVDAndro dataset, achieving an accuracy of 0.96, precision of 0.94, recall of 0.99, and an F1-score of 0.96. This combination leveraged the simplicity of FNNs and the enhanced performance of boosting techniques, proving highly effective for Android-based malware detection. The integration of LIME enabled feature-level explanation, illustrating how attributes such as application permissions or network access contributed to the model's decisions.

5) INTEGRATION OF EXPLAINABILITY TOOLS (SHAP, LIME AND OTHERS) WITH DL MODELS

The application of SHAP and LIME to ML models significantly enhances their interpretability. These tools provide both global and local insights into model predictions. This two-pronged approach offers a comprehensive view of the decision-making process. Such transparency is essential in insider threat analysis, where high levels of trust are required. For a summary of the reviewed techniques, refer to Table 5

SHAP was applied alongside DNNs trained on the UNSW-NB15 dataset. It successfully explained the model's predictions by quantifying the significance of specific features [59]. In a CNN+LSTM model on the AWID dataset, SHAP provided insights into the interaction of spatial and temporal characteristics. This contributed to a high accuracy of 0.99, despite lower precision (0.57) and recall (0.31) [39]. SHAP also facilitated interpretability in a DNN trained on the N-BaIoT dataset. It yielded excellent performance metrics, including accuracy, precision, recall, and F1 score all at or near 0.99 [59]. Additionally, SHAP proved useful in evaluating dynamic datasets such as IoT23. In one instance, SHAP enabled interpretability in a DNN that achieved 0.99 accuracy, 0.99 precision, 0.98 recall, and a 0.99 F1 score [58]. SHAP was instrumental in uncovering feature importance within the CNN+CIL model. This was particularly important during performance variations such as F1 score and AUC drops [71]. Moreover, SHAP enhanced the interpretability of hybrid models like BO-GRU-LSTM with EFFS. This model achieved an F1 score of 0.95 on the Honeypot22 dataset. SHAP also contributed to the interpretability of XENIA (FGSM+DNN+TL). It highlighted critical features influencing predictions on the MalDroid20 dataset [38], [40].

LIME, another important explainability tool, has demonstrated its effectiveness in providing localized explanations. It does this by approximating model behavior around specific predictions. In one study, a Feedforward Neural Network (FNN) combined with a binary classifier (BC) achieved strong results on the LVDAndro dataset. It reached an accuracy of 0.96, precision of 0.94, recall of 0.99, and an

TABLE 5. Compact summary of XAI techniques integrated with deep learning models for ITD.

Paper Ref.	XAI Technique(s)	Strengths	Limitations
[59]	SHAP	- Feature importance - High accuracy & interpretability	- High computation - Mostly global explanations
[39]	SHAP	- Explains CNN+LSTM - Temporal-spatial insights	- Low precision & recall - Poor generalization
[58]	SHAP	- Local/global insights - Trustworthy predictions	- Sensitive to data changes
[71]	SHAP	- Feature impact in drops	- Manual review needed
[40], [38]	SHAP	- Hybrid model interpretability - High F1 score	- Complex model structures
[78]	LIME	- Local feature insights - Explains FNN+BC	- Inconsistent results - No global view
[23]	SHAP, LIME	- Global + Local view (SHAP, LIME) - Accurate DNN (SHAP, LIME)	- High overhead (SHAP, LIME)
[72]	SHAP, LIME	- Interpretable CNN-GRU-FCN (SHAP, LIME) - Accurate traffic classification (SHAP, LIME)	- Limited generalization (SHAP, LIME)
[66]	LIME	- Explains FNN - High accuracy & F1 score	- Local-only scope
[60]	DeepLIFT	- Gradient attribution - Accurate DNN	- Baseline sensitive
[69]	Integrated Gradients, Grad-CAM	- Visual reasoning (Grad-CAM) - Numeric attribution (Integrated Gradients)	- Requires image-like data (Grad-CAM)
[68]	GNNExplainer	- Explains GNN structure	- Limited use in cybersecurity
[76]	Dalex	- Ensemble model insights	- Less clear with complexity

F1 score of 0.96 [78]. LIME was used to provide localized explanations. These explanations offered insights into the critical factors contributing to the model's high recall rate. This integration underscores the role of explainability tools such as LIME in enhancing transparency and building trust in high-performing neural networks.

In several experiments, SHAP and LIME have been used together to leverage their complementary strengths. This has resulted in a more comprehensive understanding of DL models. For example, using the UNSW-NB15 dataset, one study employed SHAP and LIME to interpret a DNN. The model achieved an overall accuracy of 0.99 and provided both global and local feature importance insights [23]. Similarly, a CNN-GRU-FCN model was applied to the CIC-Darknet-2020 dataset. It used both SHAP and LIME to reach 0.99 accuracy. These tools also provided detailed interpretability for network traffic classification tasks [72]. In intrusion detection systems, LIME has offered valuable insights into Feedforward Neural Networks trained on the NIDKaggle dataset. These predictions were explained with 0.94 accuracy and F1 score [66]. These applications demonstrate LIME's utility in enhancing transparency across a range of DL contexts.

Beyond SHAP and LIME, other explainability techniques have also emerged to address the challenges of interpreting DL models. RuleFit has been used to generate human-readable rules for DNN predictions, as demonstrated in studies using the UNSW-NB15 dataset where models achieved an accuracy of 0.99 [23], [59]. Gradient-based techniques such as DeepLIFT and Integrated Gradients have also shown effectiveness. For example, DeepLIFT was applied to analyze feature contributions in DNNs trained on the N-BaIoT dataset, achieving 0.99 accuracy while providing clear gradient-based attributions [60]. Integrated Gradients, alongside Grad-CAM, were used on CNNs trained on custom datasets. These tools provided visual explanations for malware detection and achieved an accuracy of 0.98 [69]. Additionally, emerging tools such as GNNExplainer for Graph Neural Networks [68] and Dalex for ensemble DL models [76] have further broadened the scope of model interpretability. This trend underscores the growing importance of explainability in building trust and enhancing the usability of AI systems.

E. HYBRID ML/DL APPROACHES FOR XAI-ENABLED ITD

Hybrid methods that combine ML and DL have proven to be crucial for detecting serious internal cyber threats. These approaches are particularly effective with high-dimensional datasets, as they can uncover rich patterns by leveraging the strengths of both classical ML and advanced

TABLE 6. Key insights into ML/DL and XAI integration for ITD.

Paper Ref.	AI Method(s)	Dataset(s) Used	Performance Metrics				XAI Techniques		
			Acc.	Prec.	Rec.	F1-Score	SHAP	LIME	Other(s)
[79]	DT, RF, DNN, Transformers	WUSTL-IIOT	0.99	0.99	0.99	0.99	✓	✓	×
[80]	ML/DL	×	×	×	×	×	✓	✓	DeepVis, KerasVis, Lucid, TreeInterpreter
[81]	XGBoost	DGArchive	0.94	×	×	×	✓	×	×
	MLP	Tranco	0.94	×	×	×	✓	×	×
[82]	ML	NSL-KDD	0.93	×	×	×	✓	✓	×
[56]	DT	UNSW-NB15	0.96	×	×	×	×	×	×
	MLP		0.96	×	×	×	×	✓	×
	XGBoost		0.98	×	×	×	✓	×	×
[25]	KNN	ICU-IoT-Flock	0.99	0.99	0.99	0.99	✓	✓	ELI5, IG, Shapash
	RF		0.99	0.99	0.99	0.99			
	DT		0.99	0.99	0.99	0.99			
	NB		0.95	0.96	0.95	0.95			
	SVM		0.96	0.99	0.96	0.96			
	MLP		0.95	0.96	0.95	0.95			
	ANN		0.96	0.96	0.96	0.96			
[83]	DT	LoopWater Distribution	0.93	0.88	0.93	0.75	✓	x	x
	KNN		0.85	0.78	0.62	0.63			
	SVM		0.57	0.19	0.23	0.22			
	NN		0.80	0.80	0.80	0.78			

DL architectures. However, as these models are generally complex and operate as "closed boxes," explainability tools are essential to help users and researchers better understand the decision-making process. This section discusses the integration of ML/DL techniques for explainable ITD, including data sources, evaluation metrics, and the incorporation of explainability tools such as SHAP, LIME, and others. The discussion is supported by findings from key studies, emphasizing both the opportunities and challenges in this domain, as illustrated in Table 6.

A study conducted in [79] compared DT, RF, DNN, and transformers to evaluate their effectiveness in detecting insider threats. The results demonstrated high performance across all key metrics–accuracy, precision, recall, and F1-score–each reported at 0.99, indicating the strong effectiveness of these models in identifying malicious insider activity. To reinforce the practical relevance of the proposed models, the study employed the WUSTL-IIoT dataset, which is tailored for IoT-specific network traffic. To enhance model interpretability, the experiments incorporated well-known XAI techniques such as SHAP and LIME, which provided clear explanations of model predictions, thereby improving the predictability and trustworthiness of the results. While the study highlighted the effectiveness of DL approaches, it also noted that computationally intensive models such as transformers are generally less explainable than traditional ML models like DT and RF. Therefore, it is imperative to strike a balance between the high accuracy of DL models and the interpretability of ML models, especially in sensitive domains like ITD, where explainability is essential.

To the best of our knowledge, [56] is the first study to compare DT, MLP, and XGBoost models using the UNSW-NB15 dataset, an open-source dataset frequently used in intrusion detection. The DT model achieved an accuracy of 0.96, confirming its capability to accurately classify most instances. Similarly, the Multilayer Perceptron (MLP) achieved an accuracy of 0.96, demonstrating its ability to detect complex patterns in the dataset. XGBoost outperformed both models,

TABLE 7. Compact summary of XAI techniques integrated with ML/DL models for ITD.

Paper Ref.	XAI Technique(s)	Strengths	Limitations
[82]	SHAP, LIME	- SHAP provides strong global feature insights. - LIME enables instance-level explanations.	- SHAP is computationally intensive. - LIME can produce inconsistent results across instances.
[83]	SHAP, LIME, DeepVis, KerasVis, Lucid, TreeInterpreter	- Integrates model-specific and model-agnostic XAI. (SHAP, LIME) - Visual tools improve deep learning model transparency. (DeepVis, KerasVis, Lucid)	- Complex integration of multiple tools. (All) - Interpretation consistency varies among tools. (DeepVis, KerasVis, Lucid)
[84]	SHAP	- Identifies key features in phishing detection models.	- Does not provide per-instance explanations.
[85]	SHAP, LIME	- Combines global and local interpretability. - Enhances analyst understanding and model trust.	- Not tested on hybrid model architectures.
[59]	SHAP, LIME	- Effectively explains predictions of various classifiers.	- Lacks comparative XAI evaluation across multiple datasets.
[27]	SHAP, LIME, ELI5, IG, Shapash	- Comprehensive insights on IoT behavioral data. (All) - Provides both global and local interpretability. (SHAP, LIME, IG)	- High computational cost. (SHAP, IG) - Potential redundancy among overlapping explanation methods. (SHAP, LIME, Shapash, ELI5)
[86]	SHAP	- Highlights significant session and packet-level features.	- No support for localized explanations. - Limited temporal behavior analysis.

achieving an accuracy of 0.98, indicating its sensitivity to finer decision boundaries. Although the study did not report specific values for precision, recall, or F1-score, it is notable for applying SHAP and LIME to enhance the interpretability of the MLP and XGBoost models.

The study in [25] evaluated the performance of several machine learning models–KNN, RF, DT, Naïve Bayes (NB), Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and Artificial Neural Network (ANN)–using the ICU-IoT-Flock dataset. The highest accuracy, precision, recall, and F1 scores were all 0.99. These scores were achieved by the KNN, RF, and DT models. This shows that these models are highly effective in detecting threats in IoT environments. Naïve Bayes and MLP also performed well. Each of them achieved an accuracy and F1 score of 0.95. SVM and ANN performed slightly better. Both achieved higher accuracy scores of 0.96. These tools–ELI5, Integrated Gradients, and Shapash–which provided insights into the models' decision-making processes. Explainability was emphasized as a critical factor in promoting transparency and building trust in AI-driven IoT security systems.

In [83], DT, KNN, SVM, and NN models were tested using the LoopWater Distribution dataset. The DT model performed well. It achieved an accuracy of 0.93, precision of 0.88, and recall of 0.93. However, its F1 score was slightly lower at 0.75. This suggests a small imbalance between precision and recall. KNN gave moderate results. It achieved an accuracy of 0.85, precision of 0.78, recall of 0.62, and an F1 score of 0.63. These results suggest that the model had difficulty capturing some patterns in the dataset. In contrast, the SVM model performed poorly, achieving only 0.57 accuracy and an F1 score of 0.22, highlighting its ineffectiveness for this application. The neural network demonstrated balanced performance, with accuracy, precision, and recall all at 0.80, and an F1 score of 0.78. The study also used interpretability techniques, including SHAP. These techniques helped explain how each feature contributed to the model's decisions. This made the models easier to understand and more useful for monitoring water distribution systems.

1) INTEGRATION OF EXPLAINABILITY TOOLS WITH ML/DL LEARNING MODELS

Explainability tools are essential in bridging the gap between high-performance closed-box models and their practical application in ITD. Tools such as SHAP and LIME have been widely adopted across a range of models. For example, SHAP was applied in [25], [80], and [81] to interpret predictions from models such as XGBoost, MLP, RF, and SVM. SHAP's ability to provide consistent, instance-level feature importance rankings makes it a preferred choice for high-dimensional datasets like UNSW-NB15 and ICU-IoT-Flock.

LIME, on the other hand, excels in providing localized explanations and has been used in [25] and [83] to analyze decision boundaries in simpler models such as DT and KNN. By highlighting the impact of specific features on predictions, LIME enables practitioners to identify key risk factors in datasets such as LoopWater Distribution.

Beyond SHAP and LIME, advanced visualization tools have also been employed to enhance the interpretability of DL models. DeepVis, KerasVis, and Lucid, as discussed in [80], offer insights into feature maps and activation layers, thereby aiding in the understanding of complex

TABLE 8. Comprehensive analysis of popular XAI techniques for ML and DL in ITD.

Technique	Applicable Models	Key Strengths	Limitations	Usefulness in ITD
SHAP	ML (Tree, RF, XGBoost), DL (DNN, LSTM)	- Strong theoretical foundation (Shapley values) - Consistent across models - Captures feature impact at scale	- Computationally expensive - Slow for large DL models	Ideal for detailed audit trails and feature impact analysis, especially in post hoc threat forensics.
LIME	Model-agnostic (ML/DL)	- Fast, intuitive explanations - Easy to implement - Good for on-the-fly analysis	- Unstable explanations (random perturbations) - No global view	Useful for analyzing specific insider events (e.g., suspicious login) in real-time settings.
TreeLIME	Tree-based ML models	- Tailored for decision trees - Interpretable rule-based outputs	- Limited to tree structures - Less effective for DL or complex ML	Suitable for tree-based ML pipelines used in organizational risk classification.
DeepLIFT	DL (especially DNNs)	- Tracks activation contributions - Fast for DNNs - Suitable for structured tabular input	- Requires specific model architectures - Less intuitive than SHAP/LIME	Valuable for identifying activation-based anomalies in user behavior sequences.
Grad-CAM	DL (mainly CNNs)	- Visual explanations of inputs - Good for image-like or temporal-spatial data	- Only works with convolutional layers - Not useful for tabular logs	Best for interpreting CNN-based models applied to time-series log data or sensor streams.

DL architectures. TreeInterpreter, also highlighted in [80], provides hierarchical explanations for decision trees, making it an effective tool for intuitive visualization. Tools such as ELI5, Shapash, and IG are particularly useful for IoT-specific applications [25], offering user-friendly interfaces and advanced interpretability metrics for models like RF and ANN.

The incorporation of these tools varies according to the model and dataset. For example, model-agnostic approaches like LIME are commonly used for heterogeneous datasets, whereas feature-specific techniques like SHAP are more suitable for high-dimensional data. By combining these techniques, researchers can achieve both global and local interpretability while also addressing the diverse requirements of a wide range of stakeholders.

F. COMPARATIVE ANALYSIS OF ML AND DL-BASED XAI TECHNIQUES FOR ITD

Evaluating ML and DL with XAI reveals varied effectiveness and constraints in ITD. This section highlights a comparative analysis of the two models, referencing earlier performance and interpretability findings.

1) CORE DISTINCTIONS IN XAI FOR ML AND DL MODELS

Modern ML and DL models have increasingly adopted explainable AI techniques to improve transparency and foster trust in automated predictions. However, key differences persist in their complexity, interpretability, and adaptability. When choosing between ML and DL models for ITD, organizations must balance the trade-off between model accuracy and interpretability, particularly in high-stakes cybersecurity environments.

In general, ML-based models offer greater interpretability due to their simpler structure. For instance, RF was used in [21] on the DNSLog dataset, achieving 0.96 accuracy. The integration of ANCHORS, a rule-based XAI method, provided meaningful explanations that clarified which features contributed to the model's outputs. Similarly, KNN applied to the ICU-IoT-Flock dataset in [25] reached 0.99 accuracy, and the use of Shapash and ELI5 enabled analysts to easily trace the source of anomalous behavior.

In contrast, DL models such as DNNs and CNNs offer superior performance on large-scale and high-dimensional datasets, but at the cost of increased opacity. For example, a DNN applied to the N-BaIoT dataset in [60] achieved 0.99 accuracy, yet the underlying decision-making process was more difficult to interpret. Tools such as DeepLift and Grad-CAM were employed to improve transparency by highlighting the most influential input features. While effective, these post-hoc techniques still fall short of the native interpretability seen in traditional ML approaches.

From a performance perspective, DL models generally outperform ML models on complex tasks. Studies such as [58] demonstrated that DL frameworks could achieve both 0.99 accuracy and F1 scores in detecting insider threats within IoT environments. However, ML algorithms

like XGBoost and RF remain valuable when interpretability and data efficiency are priorities. XGBoost, for instance, attained an F1 score of 0.97 in [22] on the DDoSAttack-Log dataset–competitive performance with greater ease of explanation.

Adaptability is another distinguishing factor. DL models have demonstrated the capacity to generalize to novel and evolving threats, as shown in [9], where a federated DNN achieved 0.99 precision on the UNSW-NB15 dataset. However, these models typically require larger, well-labeled datasets and are more prone to overfitting. ML models, by comparison, often perform more reliably on smaller or imbalanced datasets and are easier to retrain in low-resource settings.

In terms of transparency, DL models continue to rely heavily on advanced XAI tools to approximate interpretability. Techniques like SHAP, Grad-CAM, and RuleFit help bridge the gap between predictive performance and analyst understanding. On the other hand, the inherently transparent nature of ML models allows for direct interpretation, which is especially beneficial in applications requiring auditability and regulatory compliance.

2) COMPREHENSIVE ANALYSIS OF ML/DL BASED XAI TECHNIQUES

While the integration of explainability tools such as SHAP, LIME, DeepLIFT, and Grad-CAM with DL models has gained traction in ITD, a closer analytical examination–supported by Table 8–reveals important trade-offs that influence their effectiveness in real-world cybersecurity applications. SHAP, for instance, offers both global and local interpretability grounded in cooperative game theory, making it well-suited for deep architectures such as DNNs and LSTMs. As demonstrated in studies such as [45] and [56], SHAP enables the tracing of influential features in models trained on complex datasets like UNSW-NB15 and IoT telemetry. However, the high computational overhead associated with SHAP limits its applicability in real-time ITD scenarios, particularly when scalability is a concern.

In contrast, LIME excels in providing quick, instance-specific interpretations, as seen in its integration with MLP models for intrusion detection in [65]. Its model-agnostic design makes it highly adaptable and efficient for real-time insider event analysis. Nonetheless, LIME's reliance on perturbation sampling introduces interpretive variability across different runs, which can reduce trust in mission-critical environments. Architecture-specific methods like Grad-CAM and DeepLIFT offer additional advantages: Grad-CAM, as shown in [48], enables visual inspection of CNN feature activations–helpful for sequential or spatial data–while DeepLIFT, applied in [59], provides granular attribution scores for neurons in DNNs. Despite their value, these techniques are limited to specific neural architectures and are less effective for heterogeneous tabular logs often found in ITD datasets.

As summarized in Table 8, no single XAI method universally addresses all challenges in DL-based ITD. SHAP is ideal for comprehensive, post hoc forensic analysis; LIME provides fast, actionable insights at the local level; and Grad-CAM and DeepLIFT are well-suited for visual and activation-based interpretation in CNN and DNN models, respectively. Therefore, selecting the appropriate XAI tool should be guided by the nature of the model architecture, the data modality (e.g., tabular, time-series, graph), and operational constraints such as latency and interpretability requirements. This analytical comparison serves as a decision-support framework for practitioners and researchers aiming to deploy interpretable, trustworthy DL-based threat detection systems.

3) REAL-WORLD CHALLENGES IN XAI

While quantitative metrics such as accuracy, precision, recall, and F1-score are central to evaluating model performance in ITD, they offer only a partial view of a system's practical utility. In real-world security operations, interpretability plays a critical role in determining whether analysts can trust, understand, and act on a model's output. Despite the widespread adoption of explainable AI (XAI) tools such as SHAP, LIME, TreeLIME, DeepLIFT, and Grad-CAM numerous interpretability challenges persist when these models are deployed in operational environments.

One key issue is cognitive load, which refers to the mental effort required by human analysts to process explanations. For instance, while SHAP provides highly detailed global and local attributions for tree-based and deep models (as shown in studies [21] and [22]), the volume and granularity of the feature contributions can be overwhelming especially under time-sensitive threat response scenarios. Analysts may find it difficult to prioritize which features matter most or to relate them directly to malicious behavior patterns. This issue is compounded when explanations lack visual simplicity or alignment with cybersecurity domain heuristics.

Another concern is explanation stability and consistency. LIME, for example, is praised for generating fast, instance-specific interpretations across models such as MLPs and ensemble classifiers [24], [65], but its reliance on random perturbations introduces variability in output. Inconsistent explanations across similar inputs can erode trust in the system, especially for security analysts who rely on repeatable evidence to inform decisions. Such instability poses a barrier to deployment in high-assurance environments like critical infrastructure or national defense systems.

Moreover, interpretation granularity and model complexity often exist in tension. Tools like TreeLIME and RuleFit attempt to bridge this gap by generating rule-based or simplified explanations that mirror decision paths, as seen in the Edge-IIoT and FedIoT frameworks [9], [47]. While these models improve transparency, they may simplify complex behaviors too much, potentially omitting subtle anomaly indicators that more sophisticated models (e.g., DL-based classifiers) would capture. Thus, there is a trade-off between

retaining model fidelity and ensuring explanations remain interpretable.

A particularly critical and emerging concern is the adversarial misuse of interpretability tools. Attackers, by reverse-engineering explanations or observing patterns in SHAP or LIME outputs, may learn to manipulate input features in a way that avoids detection. Although not extensively covered in existing ITD literature, this vulnerability underscores the need for robust, privacy-preserving XAI design–especially in federated learning systems where insider data is sensitive and decentralized, such as in FedIoT [9].

Finally, domain alignment and explanation usefulness are essential factors for operational success. Security professionals often have expectations about what constitutes normal or malicious behavior. If the explanations provided by a model do not align with these expectations–or if they highlight obscure features unrelated to typical insider activity (e.g., packet metadata instead of behavioral anomalies) the explanations may be ignored altogether, even if they are technically correct. This issue has been noted in models applying Grad-CAM or DeepLIFT to DL architectures [36], [59], where insights are powerful but may lack clear semantic meaning for analysts reviewing log data or textual activity records.

III. BARRIERS AND LIMITATIONS OF ML AND DL BASED XAI TECHNIQUES FOR ITD

While ML/DL-based XAI models have shown great promise in ITD, both approaches exhibit notable limitations and barriers that hinder their full potential. These challenges span multiple domains, including scalability, adaptability, data integration, and interpretability. Understanding these limitations is essential for selecting the most appropriate approach and addressing the specific needs of cybersecurity systems. In the following, we discuss the limitations of ML/DL-based XAI models, referencing key studies from Tables 2, 4, and 6 to illustrate these challenges, as summarized in Table 9.

A. XAI LIMITATIONS IN ML BASED TECHNIQUES FOR ITD

1) SCALABILITY CHALLENGES

A significant limitation of ML-based models in ITD is scalability. Traditional ML models, such as RF and XGBoost, often struggle with large volumes of data. For example, [25] used KNN with high accuracy (0.99) in the ICU-IoT-Flock dataset for ITD. However, as data volumes increase, KNN's reliance on distance metrics makes it computationally expensive and difficult to scale. Similarly, [22] applied XGBoost in the DDoSAttackLog dataset, achieving a precision of 0.97, but encountered scalability issues when applied to larger and more diverse datasets.

In contrast to DL models, which are designed to handle large amounts of data efficiently, ML models like SVM and RF are more computationally constrained when working with high-dimensional datasets. Although XGBoost in [22] performed well, it still struggled with large, unstructured data common in cybersecurity environments. Therefore, data scalability remains a major barrier for ML models in real-world applications, particularly as organizations generate increasing volumes of behavioral, network, and communication data.

TABLE 9. Limitations of ML-based and DL-based models.

Limitation/Barrier	ML-Based Models	DL-Based Models
Scalability	Struggles with large datasets (e.g., KNN, SVM)	High computational cost, requires GPUs (e.g., DNN, CNN)
Adaptability	Limited in adapting to new attack patterns	Struggles with generalizing to unseen attack vectors
Data Integration	Difficult to combine diverse data types	Requires advanced preprocessing for integration
Imbalanced Data	Requires oversampling/undersampling	Needs large labeled datasets despite being less sensitive
Interpretability	Easier to interpret (e.g., Random Forest, XGBoost)	Requires additional XAI methods (e.g., Grad-CAM)
Hybrid Complexity	Computationally expensive, hard to interpret	Computationally expensive, hard to interpret

2) ADAPTABILITY TO EMERGING THREATS

ML models trained on historical data often struggle to adapt to novel threats. In [23], RF was applied to the UNSW-NB15 dataset, achieving high accuracy (0.99 TPR), but the model was primarily trained on past attack patterns. When new attack strategies emerge or adversaries modify their techniques, ML models may fail to detect these novel threats. As insider threats evolve over time, models must be able to generalize from historical data to future scenarios. However, ML models such as those used in [23] often require retraining with new data–an approach that is resource-intensive and may still miss emerging attack vectors that differ significantly from known patterns.

In contrast, DL models such as the DNN used in [60] are often better equipped to detect new threats due to their capacity to learn intricate patterns and representations from large volumes of data with limited labeled samples. Unlike traditional ML models, DL architectures are more adaptive and capable of distinguishing between known and unknown threat forms without frequent retraining.

3) DATA INTEGRATION COMPLEXITY

Another challenge in ML-based ITD is integrating data from multiple sources. Insider threat indicators may originate from behavioral, network, communication, and psychometric data, which vary in format and scale. Their inclusion is essential but introduces computational and calibration challenges. As discussed in [9], which deployed RuleFit with FedIoT on the UNSW-NB15 dataset, combining diverse data types

for feature extraction and fusion is both time-consuming and technically demanding. If these features are not well integrated, the resulting model may yield poor or misleading outcomes. The ML-based XAI table emphasizes the critical role of data preprocessing and integration to ensure that combined datasets are suitable for model training.

Moreover, mapping data from various sources into a format compatible with the model can require substantial effort and time, often consuming valuable computational resources. A persistent challenge is reconciling and interpreting features from disparate data sources in a coherent framework for ITD.

4) ML DATASET CHALLENGES

Imbalanced data is a key ML challenge in ITD. In datasets such as ICU-IoT-Flock [25], malicious behavior is significantly rarer than normal operations, creating a bias toward predicting the majority class. To address this, oversampling, undersampling, and anomaly detection techniques are often applied. However, if not carefully implemented, these methods can introduce biases. For instance, [22] employed XGBoost and oversampling to improve performance on DDoS data, but such techniques can lead to overfitting–especially when malicious activities are extremely infrequent.

The ML-based XAI table further highlights that managing class imbalance is a critical issue. Failure to address this challenge can result in misleading predictions and increased false positives, which is especially problematic in real-world ITD systems.

5) CHALLENGES OF HYBRID MODELS

Hybrid ML models that combine multiple algorithms to enhance performance introduce additional complexities. For example, the hybrid KNN model in [25] required significantly more resources and fine-tuning, making it less practical for real-time applications. Although hybrid models can improve accuracy, they also increase computational demands and reduce interpretability. When several algorithms are combined, the resulting model becomes harder to interpret–a major issue when XAI is essential. In regulated industries such as finance and healthcare, explainability is a fundamental requirement, and hybrid models often struggle to meet the transparency standards demanded in these sectors.

B. XAI LIMITATIONS IN DL BASED TECHNIQUES FOR ITD

1) SCALABILITY AND RESOURCE DEMANDS

While DL models, such as DNNs and CNNs, excel at handling large datasets and achieving high performance, they impose significant resource demands. For instance, [60] applied a DNN to the N-BaIoT dataset, achieving 0.99 accuracy. However, training such models is computationally expensive and requires high-performance hardware, such as GPUs–especially when working with large and high-dimensional datasets. DL models tend to perform well when ample computational resources are available, but this becomes a barrier in resource-constrained environments, where limitations in data storage and processing power can restrict deployment.

Although ML models are generally more scalable in terms of computational cost, DL models outperform them in resource-rich environments, particularly when dealing with complex data structures.

2) LACK OF INTERPRETABILITY

A major challenge associated with DL-based XAI is the lack of interpretability. Unlike simpler ML models, where decision trees or rule-based outputs can directly support human reasoning, DL models require advanced XAI techniques–such as Grad-CAM [69] and DeepLift [60]–to provide insights into their decision-making processes. Despite these techniques, DL models are still regarded as ''closed-box'' systems, in which explaining the rationale behind specific predictions can be difficult.

For example, the DeepLift approach used in [60] to interpret DNN predictions on the N-BaIoT dataset achieved high performance, but relied on complex visualizations to convey model reasoning. This reduced transparency presents a challenge in environments where decisions must be justified to human stakeholders. In contrast, ML models–such as the RF implementation in [23]–are more interpretable, especially when combined with XAI tools like LIME or SHAP.

3) DEPENDENCY ON LARGE DATASETS

DL models are highly dependent on large, labeled datasets to perform optimally–posing a significant challenge in domains where labeled data is scarce. The DNN in [60], which achieved 0.99 precision on the N-BaIoT dataset, would likely exhibit degraded performance in low-data scenarios. Unlike ML models, which can be effective with smaller datasets or can leverage unsupervised learning approaches, DL models require extensive labeled data to detect complex patterns and generalize effectively to unseen threats. This is a major limitation in cybersecurity, where labeled data for emerging insider threats is often unavailable.

IV. FUTURE RESEARCH DIRECTIONS

As ITD continues to evolve, future research is expected to focus on several key areas aimed at enhancing the effectiveness, scalability, and interpretability of models. These advancements will be driven by the integration of more advanced AI and ML techniques, including ML models, natural language processing (NLP), and privacy-preserving methods. These developments will improve the ability to identify insider threats while addressing ethical concerns and reducing the lack of confidence in automated decision-making processes.

A. INTEGRATION OF ADVANCED AI TECHNIQUES

From the perspective of ITD, the future will be inclined toward the inclusion of even more advanced artificial intelligence techniques to process diverse data sources in

real time. For example, state-of-the-art NLP models such as BERT and Transformer, which have proven particularly effective in handling unstructured data (e.g., emails, chat logs, and social networks), are expected to see increased adoption. More specifically, [30] describes how NLP can be used to identify multi-layered behavioral patterns that may indicate insider threats, leveraging communication data as one of the most promising sources for detecting changes in user behavior.

Furthermore, there will be continued advancements in ML models, particularly recurrent neural networks (RNNs) and long short-term memory networks (LSTMs), as used in [60] to analyze network logs and user activity to identify potential intrusion patterns that other ML models, such as XGBoost [22], may not be able to capture. It is anticipated that future research will involve combining the results of these DL models with more complex algorithms to effectively identify sophisticated insider threats through the analysis of historical activity and the prediction of future malicious behavior. These developments in AI and ML will enable ITD models to more accurately capture dynamic behavioral patterns and generate more reliable predictions, even in complex environments with vast amounts of unstructured data.

B. REAL-TIME DATA PROCESSING AND MULTI-MODAL MODELS

A significant future direction in ITD will be the development of real-time processing systems capable of handling multi-modal data. Insider threats often involve a combination of behavioral data, network activity logs, communication patterns, and device usage data. Traditional ML and DL models, including RF [23] and CNNs [72], have demonstrated strong capabilities in analyzing specific types of data, but the integration of these data sources in real time has not yet been fully realized.

Future models will need to process these different data types simultaneously, necessitating the development of multi-modal architectures capable of combining structured data (e.g., network logs) with unstructured data (e.g., textual information). This can be achieved through multi-input DL networks or graph neural networks (GNNs), which represent relationships between entities (e.g., users, devices, and actions) as graphs. Research such as [9], which employed RuleFit in FedIoT to integrate heterogeneous data sources, demonstrates the promise of hybrid models that combine diverse input types. Future efforts will likely focus on real-time processing of these multi-modal data streams, thereby enhancing the model's ability to detect insider threats as they emerge, rather than relying solely on historical data.

C. PRIVACY-PRESERVING TECHNIQUES

As concerns about user privacy and data protection grow, future ITD models will need to incorporate privacy-preserving techniques. Federated learning is one such approach that enables models to be trained on decentralized data without sharing sensitive information between different parties, as highlighted by [25] in their work on privacy-conscious models for IoT environments. This technique is particularly valuable in ITD, as it allows organizations to build models that respect user privacy while still benefiting from large, distributed datasets.

Furthermore, differential privacy techniques will become increasingly important for ensuring that personal data is not exposed during the training process. These methods will enable models to analyze behavioral and network activity without compromising individual privacy, thereby addressing ethical concerns related to surveillance and user data monitoring. Such privacy-preserving approaches will facilitate the development of secure and compliant ITD systems–an increasingly critical requirement as regulatory frameworks (such as GDPR) become more stringent. Future research will likely focus on integrating federated learning and differential privacy with both traditional and DL models, creating a balance between effective threat detection and the preservation of data privacy.

D. ADVANCEMENTS IN EVALUATION METRICS FOR XAI

As the complexity of ITD models increases, particularly with the integration of DL, the demand for effective evaluation metrics for XAI will become more pronounced. Current metrics primarily focus on traditional performance indicators such as accuracy, precision, and recall. However, these metrics do not account for the explainability of the model, which is crucial in cybersecurity, where understanding the reasoning behind model decisions is essential for trust and regulatory compliance.

Future research will likely focus on the development of standardized evaluation metrics that balance performance with explainability. Fidelity, which measures how well an explanation aligns with the model's actual predictions, and coverage, which assesses how much of the model's behavior is captured by the explanation, will become essential metrics. [30] emphasizes the importance of these metrics in ensuring that explanations are not only accurate but also practical for cybersecurity analysts.

E. BENCHMARK DATASETS AND CONSISTENT EVALUATION FRAMEWORKS

The ITD system has remained an area of active research and development due to various factors, including the lack of benchmark datasets used to assess the effectiveness of different XAI methods. Future advancements will likely involve the creation of datasets that are as realistic and representative of actual insider threats as possible. These datasets will be essential for evaluating the generalizability of models and assessing their ability to handle different categories of threats, such as advanced persistent threats (APTs) and zero-day attacks.

Furthermore, there is a growing need for standardized benchmarking criteria that would enable fair comparisons between different models and XAI approaches. The

development of such frameworks will allow researchers to evaluate model performance against common standards and datasets, leading to more reliable and comparable outcomes. The use of benchmark datasets that maintain reasonable fidelity and coverage, along with user trust metrics, will help ensure that the models are accurate, explainable, and practically applicable in cybersecurity operations.

F. EXPLAINABLE DEEP LEARNING MODELS

The future scope of research in ITD lies in improving the interpretability of DL models. Despite the high performance of both DNNs and CNNs in threat detection, the issue of interpretability remains a significant challenge. Methods such as Grad-CAM [69] and DeepLift [60] offer some degree of interpretability; however, achieving interpretability with such approaches can impact the performance of DL models. Future studies are expected to explore novel techniques for describing complex DL architectures, including approaches such as attention mechanisms that highlight critical features or regions of the input data. These advancements will make DL models not only more interpretable but also more applicable in cybersecurity domains, where understanding model reasoning is essential.

V. CONCLUSION

This survey has analyzed how the XAI techniques contribute to detecting malicious insiders. Insider threats are a significant issue in the cybersecurity domain, and XAI is used to improve both the performance and reliability of the detection of these threats. By applying the two most common approaches as ML and DL models, substantial advancements have been made not only in detecting possible threats and dangers but also reporting the rationale behind such analytical results that would help security analysts to comprehend and analyze the model's results.

In this context, the integration of XAI techniques provides a powerful solution by making AI-based threat detection systems more interpretable and actionable. Techniques such as SHAP, LIME, RuleFit, and DeepLIFT offer local and global explanations that help analysts understand which features influence a model's predictions. Through a comprehensive review of supervised, unsupervised, and hybrid ML/DL models used in ITD, this paper has highlighted how XAI methods improve not only transparency but also usability and regulatory compliance.

The survey also identified several core challenges in this space. These include high false positive rates, lack of contextual understanding, scalability issues, and limited support for explainability in complex or real-time systems. While ML models generally provide better interpretability, they may struggle with scalability and adaptability. In contrast, DL models excel at recognizing complex patterns but often require significant computational resources and lack transparency. XAI offers a unifying solution by improving the interpretability of both ML and DL systems, making them more suitable for deployment in real-world cybersecurity environments.

Looking ahead, future research should prioritize the development of privacy-preserving and federated learning techniques that can support explainable threat detection in decentralized and sensitive environments. There is also a pressing need for XAI methods capable of handling heterogeneous, real-time data streams, particularly as insider behaviors span multiple modalities and contexts. Moreover, establishing standardized evaluation frameworks that incorporate not only model performance but also interpretability, explanation fidelity, and human usability will be vital for the practical adoption of these systems.

In conclusion, this survey underscores the essential role of XAI in advancing ITD. As organizations face increasingly sophisticated threats, building intelligent systems that are not only accurate but also interpretable and trustworthy will be key to enabling proactive and informed cybersecurity responses.

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